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Key Points:

- We developed a deep-learning tool to classify the crater geomorphological type (ghost, layered, secondary and regular) from an image
- The tool reach a global classification accuracy of 81% on a data set covering the entire surface of Mars
- Automatic classification enables high-speed analysis, making easier crater-based surface age estimation for areas with very small craters

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Automatic Crater Classification Using a Deep-Learning-Based Pipeline

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Abstract Identifying and classifying impact craters on Mars is crucial for understanding the planet's geological history and surface evolution. Traditional crater classification relies on manual annotation methods, which are often limited by human biases and the difficulty of interpreting geomorphological features. Classifying crater is both necessary and challenging. This task is necessary because planetary surfaces ages are estimated by crater density, but some crater types have to be discarded from the analysis to avoid large estimation errors. For instance, secondary impact craters or buried craters should not be counted for a given surface unit. This task is challenging because the different crater classes share a lot of common features. In this study, we present a deep-learning approach for automated crater classification on Mars using a YOLOv11 architecture. Our method is trained on downsampled (50 m/pixel) Context Camera images (CTX camera) and a human-annotated crater database of craters ≥1 km in diameter. This pipeline is one of the first automated tools capable of classifying craters ≤1 km, addressing a critical gap in high-resolution analyses. Validated across diverse Martian terrains, it shows potential for planetary dating and comparative geology. Incorporating a pre-processing step to reduce false positives, we train and test our model globally on separate data sets. Our results demonstrate 81% accuracy, providing a reliable tool for large-scale planetary surface analysis. This pipeline advances planetary geosciences by enabling systematic, feature-based crater classification and highlights deep learning's potential in planetary exploration.

Plain Language Summary In this work, we developed an AI-based tool to assist planetary scientists in classifying impact craters on Mars. Craters provide valuable insights into the age of planetary surfaces; however, not all craters are suitable for dating, as some are secondary or predate the current surface. Traditionally, distinguishing between crater types has been done manually—a process, that is, both time-consuming and prone to human error. Our approach leverages a deep learning model trained on 50 m per pixel Martian imagery to automatically classify each crater into one of four categories. The model was evaluated on unseen data from various regions of Mars and achieved an accuracy of 81%. Additionally, we enhanced the spatial accuracy of thousands of craters using a shape-detection algorithm, improving the reliability of the data set. This tool offers a more efficient and objective method for crater analysis, ultimately contributing to more accurate surface age estimations. Beyond Mars, our approach may be used for automated geological mapping on other planetary bodies, such as the Moon or icy moons like Europa, with further model adaptation and training.

1. Introduction

Impact craters are among the most prominent geomorphological features found on solid planetary surfaces (Leighton et al., 1965). Their morphology, spatial distribution, and degradation states provide crucial insights into the geological history, surface processes, and relative and absolute age of planetary terrains (Hartmann & Neukum, 2001; Neukum et al., 2001). Moreover, crater counting is the only dating method available for planetary surfaces for which a sample return has not yet been achieved. So far, only a few locations on the Moon and two asteroids can be absolutely dated using the radiochronology method. Mars, on the other hand, has been extensively dated by the crate-density method (Benedix et al., 2020; Neukum & Horn, 1976). Over the past decades, various automatic crater detection algorithms have been proposed, particularly for Mars. These approaches typically fall into three categories: (a) template matching methods (Salamuniccar & Loncaric, 2010), (b) digital terrain model (DTM)-based algorithms using topographic derivatives (Chen et al., 2018; Lee & Hogan, 2021; Tao et al., 2021; Zuo et al., 2016), and (c) modern machine-learning or deep-learning methods (Benedix et al., 2020; La Grassa et al., 2023; Martinez et al., 2025b). Machine learning techniques can also be used to detect other geomorphological features such as ice sheet (Su et al., 2023), pitted cones (Mills et al., 2024) or rockfalls (Bickel et al., 2020). Recently, the application of convolutional neural networks (CNNs) and object detection models has significantly improved crater detection accuracy, particularly when leveraging high-resolution data sets such as CTX V01 Mosaic which provides a 5 m/pixel resolution.

Despite these advances in crater detection, the classification of craters based on their morphological state or origin remains a relatively unexplored area. A notable exception is the work by Lagain et al. (2021), who manually classified more than 376,000 Martian craters larger than 1 km into four categories: *regular*, *layered*, *ghost*, and *secondary* craters. However, this classification relied entirely on manual inspection, which is inherently time-consuming, subject to human bias, and difficult to scale. Moreover, studies such as (See et al., 1995) have demonstrated that repetitive tasks like manual annotation lead to performance degradation after only short periods of concentration.

While the (Lagain et al., 2021) database provides a comprehensive classification of Martian craters larger than 1 km, its manual annotation process inherently limits its scalability and resolution. A key novelty of our deep-learning approach lies in its potential applicability to craters smaller than 1 km, a size range that remains unexplored in automated classification due to the large number of craters which makes the work harder and harder as much as you want to classify small craters.

In this study, we introduce the first fully-automated crater classification pipeline based on deep learning, capable of distinguishing crater morphologies using high-resolution CTX imagery. Our approach is built upon the YOLOv11 architecture—a state-of-the-art object detection framework—trained to assign each crater into one of four classes (Lagain et al., 2021): regular, ghost, layered, and secondary. By incorporating a pre-processing step based on Hough circle detection (Duda & Hart, 1972) to refine crater bounding boxes, our method enhances the alignment between the visual features and the label annotations, reducing misclassifications.

By leveraging the high spatial resolution of the CTX V01 Mosaic (5 m/px) and the robustness of the YOLOv11 architecture, our pipeline should be able to extend classification capabilities to sub-kilometer craters, thereby addressing a critical gap in planetary geomorphology. This advancement not only enhances the precision of crater-based surface dating but also enables the analysis of geological processes acting at finer scales, such as secondary cratering and local resurfacing events.

To the best of our knowledge, this is the first attempt to classify Martian craters using a deep-learning approach, moving beyond detection to enable automated geomorphological interpretation at the planetary scale. Here, we develop a method to only classify the crater and decide not to combine classification and detection to simplify the problem. We evaluate our model's performance using a test set distributed across various quadrangles of Mars at global scale, providing insights into regional variations in classification accuracy. Furthermore, we discuss how such classification could be extended to other planetary bodies and ultimately contribute to more accurate surface dating by filtering out misleading features such as secondary and degraded craters.

2. Data

2.1. The Most Complete Classified Crater Database on Mars

The Mars Crater Database presented by Lagain et al. (2021) provides one of the most comprehensive classifications of Martian craters to date. Using imagery predominantly from THEMIS mosaics at a resolution of 100 m/px, this crater database classifies 376,419 Martian craters larger than 1 km in diameter into four distinct categories: *Ghost*, *Layered*, *Secondary* and *Regular* (noted as *Valid* in the initial database) craters. It comprises respectively 24,530 *Ghost* craters (6.5%), 8,445 *Layered* craters (2.3%), 55,309 *Secondary* craters (14.7%) and 288,155 *Regular* craters (76.5%) and highlighting the diverse morphologies present on the Martian surface. In this study, we adopt the terminology "*regular*" to refer to ordinary craters, consistent with their classification. In this work there is no question about the validity of the crater (detected or not): all objects are truly craters and the purpose of the tool is to classify them. This comprehensive classification facilitates refined analyses, such as surface age dating, by clearly delineating morphological characteristics across a wide range of crater sizes and types.

Each of the crater categories present in the Lagain et al. (2021) is further defined below:

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1. Ghost craters: Primary craters that are characterized by significant degradation and are partially or completely buried, distinguishing them from eroded craters (R. A. Craddock, 1997). Importantly, as secondary craters, they shall not be considered in dating processes, as their formation predates the surface being dated.
2. Layered: Primary impact craters, which present a continuous rampart of ejecta blanket. They are characterized by fluidized ejecta patterns around the crater, suggesting that the material had a mud-like flow during impact (Barlow et al., 2000; Costard, 1989).
3. Secondary: Crater formed by the ejecta of a primary impact event. Secondary craters tend to be aligned radially around a primary crater. They are characterized by a much smaller size than the primary crater. Their shape can be more irregular compared to primary craters because of lower impact velocities. Several studies have explored this subject (Quantin et al., 2016; Robbins & Hynek, 2011a, 2011b). The authors state that secondary craters should not be considered in the dating process, as their formation is not directly related to the primary impact flux and can lead to an overestimation of surface ages.
4. Regular: Craters categorized as regular are the ones which could not be assigned to any of the three previous categories (assigned as "Valid" in the initial Lagain database).

It is important to note that a minimum crater size was arbitrarily set through manual inspection (Lagain et al., 2021). Consequently, threshold issues may arise near the 1 km limit. Because the labeling was done manually, some craters slightly larger than 1 km may be missing from the database, while others slightly smaller than 1 km might be included. To address this issue, we filter the database to only keep craters with diameters greater than 1.2 km.

2.1.1. Other Possible Crater Classes

Another common morphological classification which can be made is the distinction between *simple* and *complex* craters based on their structural features. Complex craters are generally defined by the presence of central peaks, terraced walls, or flat floors, while simple craters lack these features and typically exhibit a typical bowl-shaped morphology (Baker et al., 2011; Cintala, 1977; Cintala & Grieve, 1998; Kalynn et al., 2013; Pike, 1980; Robbins & Hynek, 2012).

In this study, we deliberately chose not to include this distinction. Indeed, identifying simple and complex craters does not necessitate sophisticated deep-learning techniques. Furthermore, complex craters are not numerous since only the largest craters have these characteristics (Chandnani et al., 2019; Herrick & Lyons, 1998). They may very easily manually checked by geomorphologist. On Mars, the transition between simple and complex craters occurs at diameters between about 3 and 8 km (Croft, 1985; Pike, 1980). Furthermore, integrating such a classification would add unnecessary complexity, as craters categorized as *Regular* or *Layered* could simultaneously belong to either the simple or complex morphological types, complicating the training and evaluation of our deep-learning model, while being useless for dating purposes.

2.2. CTX V01: A New Global Mosaic

The CTX V01 Mosaic is the latest version of the global mosaic compiled by the Bruce Murray laboratory from images captured by the CTX instrument aboard the Mars Reconnaissance Orbiter (MRO) (Dickson et al., 2024; Malin et al., 2007) https://murray-lab.caltech.edu/CTX/tiles/. It encompasses more than 100,000 individual panchromatic image patches, which have been combined to create a mosaic with a total size of 5.7 trillion pixels, covering 99.5% of Mars from 88°N to 88°S. For practical reasons, the Bruce Murray Laboratory team reprojected the mosaic into an equirectangular projection and divided it into 3,960 tiles, each covering an area of $4° \times 4°$.

For the purpose of this study and the previous one about crater detection (Martinez et al., 2025b), we excluded the polar ice sheets and seasonal caps by using only the tiles between 80°N and 80°S. Additionally, since the smallest crater in the database is 1 km in diameter, we automatically downsampled the tiles to a resolution of 50 m/px to avoid detecting craters that smaller than the ground truth database (1 km in diameter).

2.2.1. Quadrangle Repartition

At this stage, we had 3,600 image tiles at a resolution of 50 m/px. The next step was to ensure that all craters retained their circular shape. Indeed, with the native projection of the mosaic, the higher the latitude is, the more craters will appear with an oval shape. To correct this distortion, we reprojected the tiles—as done in our previous

Table 1
Definition of the Quadrangle as a Function of Latitude (Applied to Both North and South Latitudes)

Latitude range	Quadrangle size	Surface [Km^2]
80°–64°	16° × 40°	689,884
64°–52°	12° × 20°	444,276
52°–40°	12° × 12°	349,434
40°–32°	8° × 8°	181,055
32°–24°	8° × 8°	197,601
24°–16°	8° × 8°	210,300
16°–8°	8° × 8°	218,906
8°–0°	8° × 8°	223,251

Note. The surface represent the image surface once projected in local stereographic projection, to avoid projection distortion.

study (Martinez et al., 2025b)—into a local stereographic projection centered at the midpoint of each tile. One issue encountered during this processing step is that a 4° × 4° tile in the polar regions covers a significantly smaller surface area than at the equator. Also the numerous tiles boundaries in the polar areas could create difficulties when learning and merging all results. In order to avoid this effect, we decided to merge adjacent tiles in the mid/high latitude regions to create larger "quadrangles," ensuring a more consistent coverage. Table 1 shows the quadrangle size and surface with respect to the latitude range; Figure 1 shows the selected quadrangle grid merged from CTX Tiles.

The next step was to split our global, reprojected data set into three distinct subsets. As is standard in deep learning workflows, we required one subset for training, another for validation (to monitor the training process), and a third for testing, which is used exclusively to evaluate the model's final performance. The latter is never used during training and is therefore ideally suited for a fair and unbiased assessment. We divided the data set into training (70%), validation (15%), and test (15%) subsets. The partitioning was performed randomly for the training and validation subsets, ensuring that latitude, illumination, and geological diversity were equally represented, following a stratified approach. The test subset was also composed of randomly selected tiles, but not exclusively: we deliberately included tiles that have been extensively studied in the literature, most of which cover well-known landing sites and thus serve as valuable reference areas for comparison (Hynek & Di Achille, 2017; Parker et al., 2010; Quantin-Nataf et al., 2021).

3. Pre-Processing

3.1. Crater Bounding-Box Improvement Using Hough Circle Detection

In order to verify the precision of the ground truth database, we examined several images. As seen in Figure 2, the localization and radius of craters which are registered in the ground truth database (green circles) do not perfectly overlap the crater seen in the image. We observe that the craters are often shifted in an apparently random direction and that this difference becomes more significant as the size of the crater. One plausible explanation for these shifts is the errors and uncertainties on the initial THEMIS mosaic data set in comparison with our CTX data set. Another possibility may be uncertainties in the crater position and size from Lagain et al. (2021), initially from Robbins and Hynek (2012), simply because they were done manually.

In Appendix A, we propose a pre-processing step based on the Hough algorithm (Duda & Hart, 1972; Hough, 1962) that improves the match between the crater database center and size and the actual image. This step is designed to build a more precise ground truth database, which will enable us to achieve the goal of 0.77 in accuracy obtain by ultralytics in their classification performance benchmark on ImageNet Large Scale Visual Recognition Challenge (https://docs.ultralytics.com/fr/tasks/classify/#models).

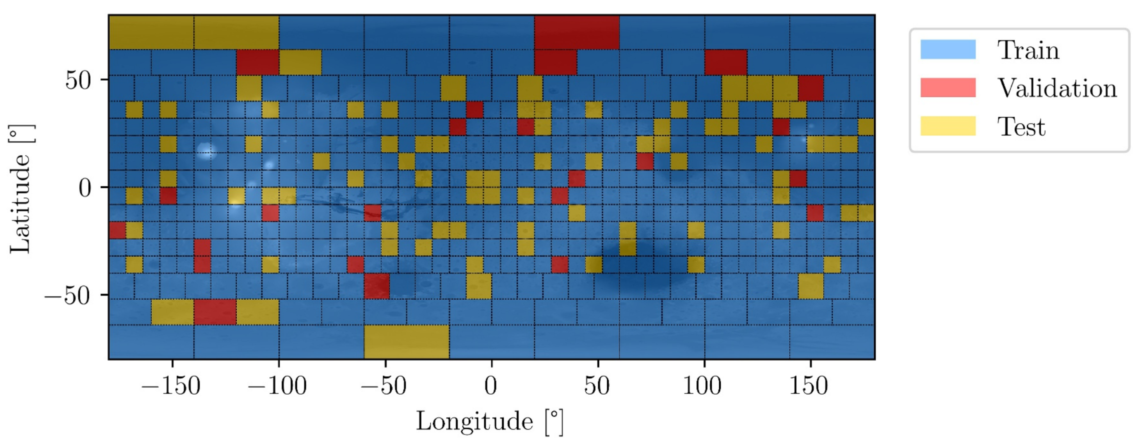

Figure 1. Definition of the quadrangle and their attribution for the Train, Validation and Test.

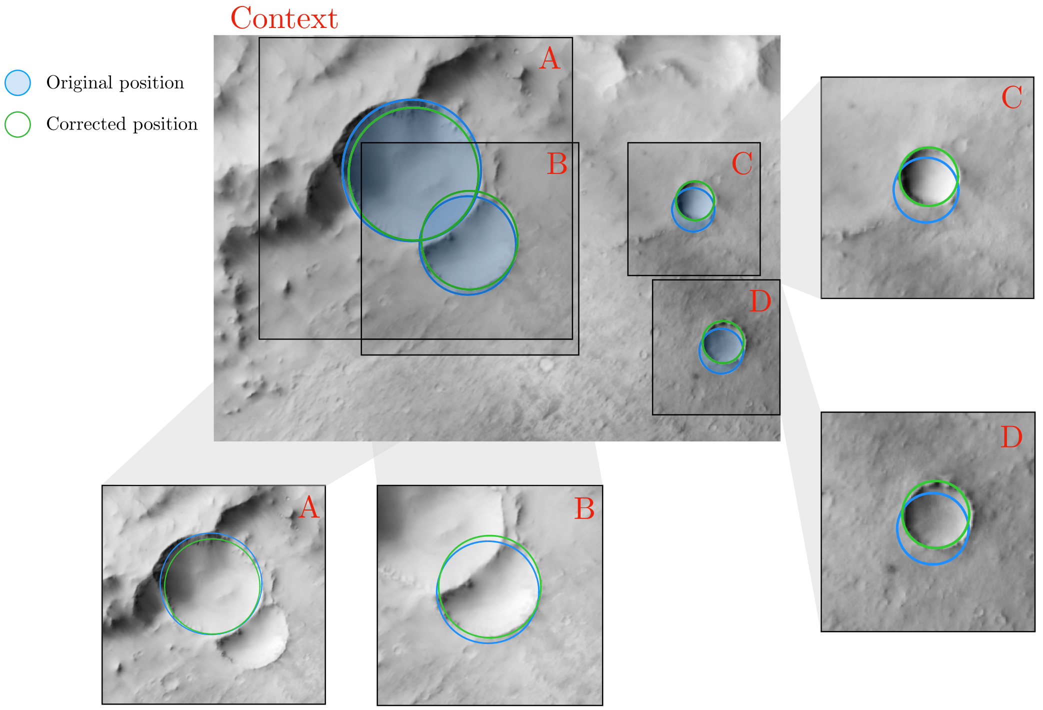

Figure 2. Initial crater database (Lagain et al., 2021) in green superposed with CTX images located around 0.16°E, 12.2°S on Mars. The position and radius are nearly correct but an improvement is required. The more craters appear as big, the less the correction is needed. The original position appear in green and the improved position appear in blue.

3.2. Data Preparation

For each crater listed in the ground truth database, we extracted a square image patch from the corresponding CTX tile. To determine the size of each patch, we set a buffer to the crater radius equal to 2.5 radius. In other word, each image patch have a size of five times the crater radius and centered on the crater's coordinates. These patches were then saved into folders corresponding to their crater class.

3.2.1. Sub-Data Set: Balanced Crater Class and Size Sampling

Given the extensive size of the global crater database (376,419 entries) and the highly heterogeneous crater repartition across the different morphological classes (as detailed in Section 2.1), we constructed a balanced sub-data set to ensure equitable representation of each class. To enhance the robustness of the learning process, we selected a subset (which we call called the "subdataset") comprising approximately 7% of the total data set, with each class represented in equal proportions (see Table 2). Table 2 illustrates the total number of crater image patches we have in the training, test and validation databases for each category and in total. This sampling allowed for a more efficient training phase and ensured that each class was equally represented to avoid class imbalance

Table 2
Total Number of Crater Images in the Initial Database and in Our Subdataset Used to Train, Evaluate and Test the Model

	Ghost	Layered	Secondary	Regular	Total	Total augmented
Lagain et al. (2021)	24,530	8,445	55,309	288,155	376,419	–
Train	6,715	6,715	6,715	6,715	26,860	214,624
Validation	478	478	478	478	1,912	15,280
Test (from Lagain et al., 2021)	3,334	1,249	7,842	39,389	51,814	–

Note. The train, validation and test cases are selected from quadrangled defined in Figure 1.

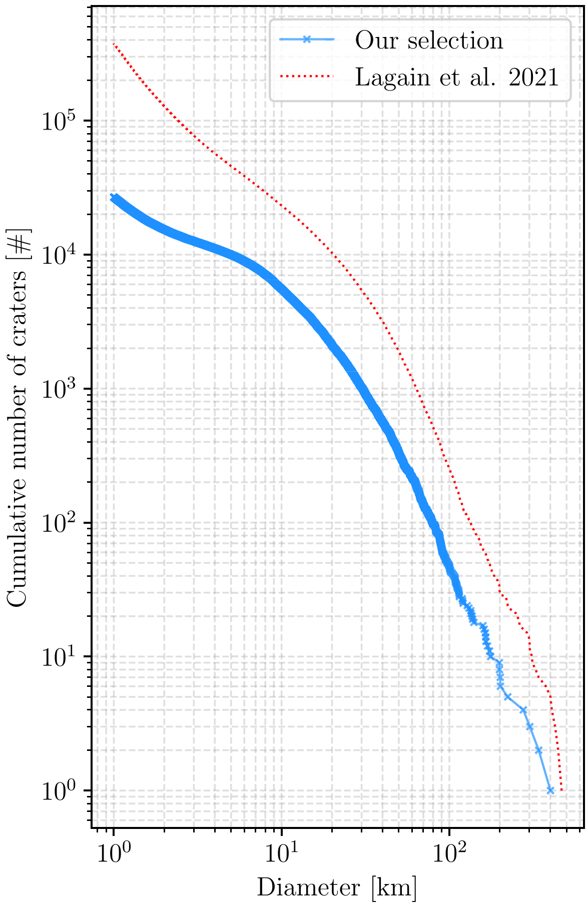

Figure 3. Cumulative Crater Size Frequency Distribution (CSFD) of our training data set compared to the full crater database. The trend is similar with a slight bias toward the larger crater.

during model learning (Johnson & Khoshgoftaar, 2019). This sampling strategy significantly improved training efficiency while preserving diversity across classes. However, due to the strict requirement of balancing the data set across all classes, the requested number of craters per class occasionally exceeded the available entries in the original data set—especially for *Layered* crater classes. Also, some craters near the border of the quadrangles are removed. As a result, the final proportions may slightly differ from the ideal theoretical distribution. Nevertheless, our selected subdataset maintains a realistic distribution of crater sizes, as confirmed by its alignment with the Crater Size Frequency Distribution (CSFD), shown in Figure 3.

Given the necessity to limit our training data set to a balanced subdataset where each crater class is equally represented, we now need to expand its effective size by using geometric properties of the craters. In order to be able to use a sufficiently large model while maintaining high accuracy and reducing the risk of overfitting, we implemented two systematic data augmentation strategies:

3.2.2. Data Augmentation to Increase the Total Number of Images

For each image patch, we generated seven additional versions of it by applying a series of geometric transformations designed to preserve the intrinsic properties of craters. Specifically, we utilized rotations of 90°, 180°, and 270°, along with horizontal and vertical flips applied to both the original image and its 90°-rotated counterpart. These transformations were selected based on the circular symmetry of craters, ensuring that essential morphological features were conserved while introducing variability in illumination and spatial orientation. With this approach referenced in Table 2, this augmentation process yields a total of eight distinct images per original patch—effectively expanding the data set size by a factor of eight and facilitating the training of deeper, more robust classification models.

This augmentation strategy was deliberately limited to this complete but minimal set, as it already covers all combinations of rotations and flips, making further transformations redundant. It was only also done for the train and validation data set since we wanted the test data set to obey to realistic condition so the test data set encompass all craters of the Lagain et al. (2021) data set present in the test area.

3.2.3. Data Augmentation to Increase Robustness

The YOLO pipeline includes a comprehensive series of data transformations that are crucial for enhancing both the robustness and performance of the model. These transformations, which is extremely common in IA processes (Krizhevsky et al., 2012; Shorten & Khoshgoftaar, 2019; Taylor & Nitschke, 2018), introduce variability into the training data, allowing the model to better generalize to unseen examples and reducing the risk of overfitting.

To build a more reliable classifier, we applied a diverse set of augmentations that simulate various image dynamics, such as changes in contrast, brightness, and noise. These augmentations help the model learn more invariant features and improve its ability to handle real-world variations.

For example, real images may contain pixels flagged as "no data," often resulting from gaps in observation during the mosaicking process. To account for this and train a model resilient to such artifacts, we introduced a data augmentation step that randomly adds black patches to training images, simulating missing data in a controlled way.

3.3. System Setup

In this study, we parallelized the training on 8 NVIDIA GeForce GTX TITAN X GPU boards. The GPUs are installed within a single server equipped with two Intel Xeon E5-2690 v4 @ 2.60 GHz 28 core processors and

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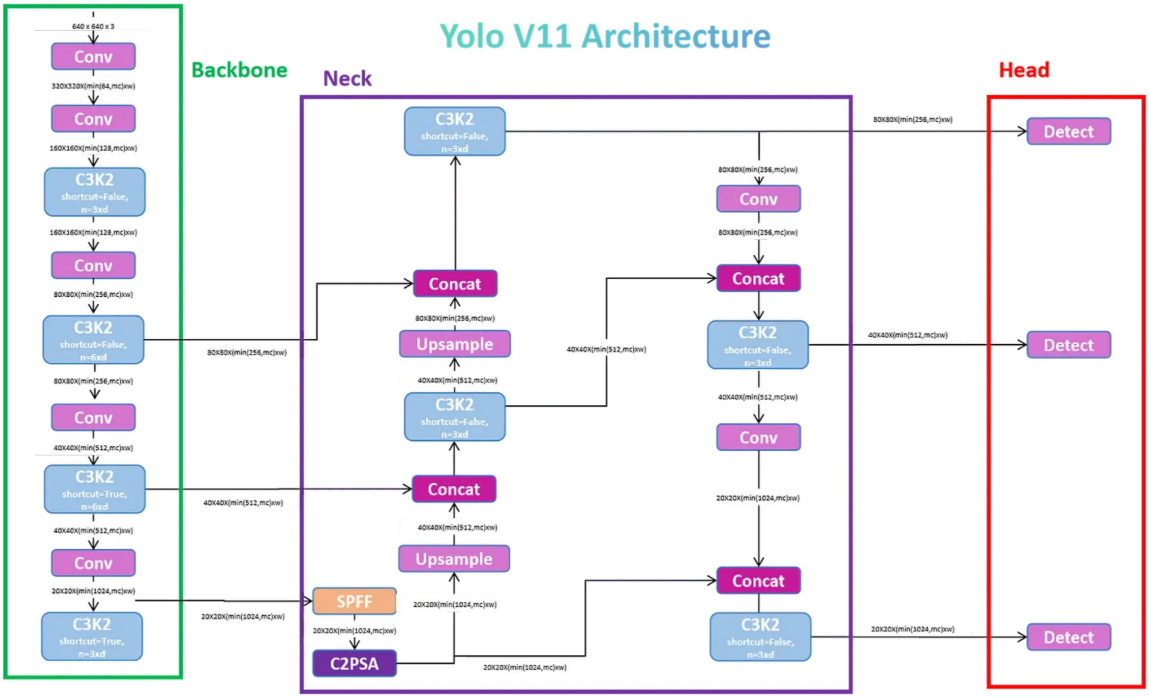

Figure 4. Yolov11 architecture as presented by S. Nikhileswara Rao on https://medium.com/@nikhil-rao-20/yolov11-explained-next-level-object-detection-with-enhanced-speed-and-accuracy-2dbe2d376f71 and as resume from Khanam and Hussain (2024).

756 GB of RAM. The server ran Ubuntu 20.04.6 LTS. A Python environment containing TensorBoard (version 2.17.0), PyTorch (Paszke et al., 2019) (version 1.13.1), and Ultralytics (2025) (version 8.3.79) was used to implement, train, and visualize the networks. With our configuration, the pre-processing step (including image cropping and reprojection across the entire data set) took approximately 30 min when parallelized on 55 CPUs. Training typically required 5 min per epoch, processing 1,677 training batches and 478 validation batches, each comprising 128 images, distributed across 8 GPUs in parallel (see Figures 5–7 for learning rate, the loss evolution and validation accuracy, respectively). For inference, the process achieved latencies of 12.2 ms per frame on the CPU and 2.6 ms per frame on the GPU.

4. Method

4.1. Classifier

We chose to use the classification model of the YOLO-v11 medium-sized architecture. Yolov11 is the latest model in the YOLO series. It was presented in September 2024 at the YOLO Vision conference YV24. The network architecture, presented in Figure 4, introduces a number of improvements compared to previous versions such as C3K2 blocks (Cross Stage Partial with kernel size 2), SPFF (Spatial Pyramid Pooling Fast), and advanced attention mechanisms like C2PSA (Convolutional block with Parallel Spatial Attention) which contribute to improved model performances in several ways, such as enhanced feature extraction (Khanam & Hussain, 2024).

The loss function used in the YOLOv11 model in the classification task is a cross entropy loss function. This component measures the divergence between predicted class probabilities and the ground truth labels. It is based on cross-entropy principles, which are effective for classification tasks by penalizing incorrect class predictions and rewarding correct ones (He et al., 2024).

At inference, this classifier proposes a score for each class and the class is attributed to the highest class score.

Several studies have demonstrated that while increasing the model size beyond the medium variant can lead to marginal gains in precision, these gains come at a significant computational cost. For instance, Jegham et al. (2025) showed that larger models yield diminishing returns in accuracy improvements compared to the substantial increase in inference time and resource consumption. Consequently, we determined empirically that

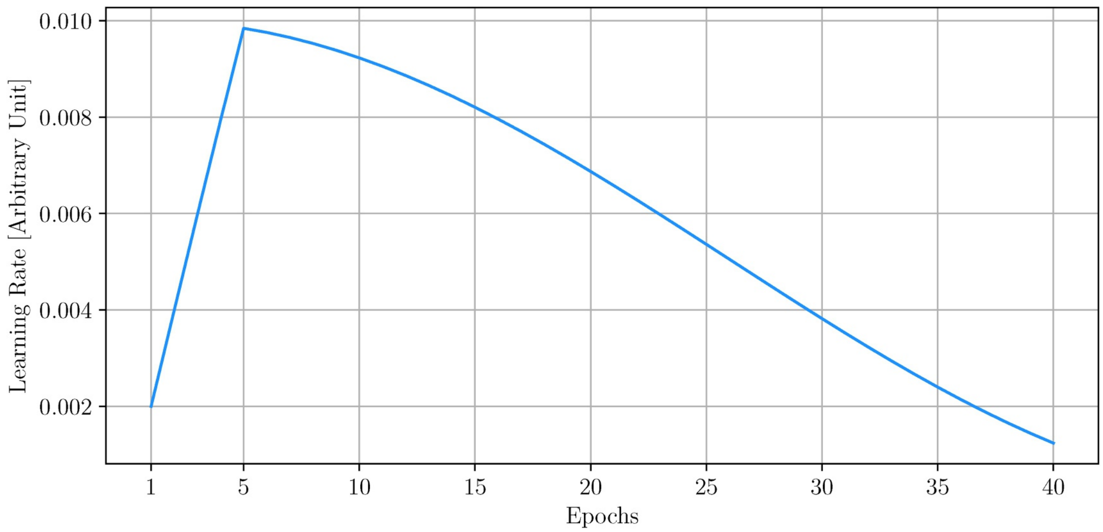

Figure 5. Learning rate variations used during training.

the medium model provides the best trade-off between accuracy and speed, making it ideally suited for our application.

These improvements are particularly relevant for crater classification, where objects are small, low-contrast, and morphologically diverse. The enhanced feature fusion and attention mechanisms of YOLOv11 allow the model to better capture subtle rim and ejecta morphologies, while the multi-scale detection architecture ensures robustness across a wide range of crater sizes and illumination conditions. Furthermore, the improved training stability and regularization strategies make YOLOv11 well suited to the limited and heterogeneous labeled data sets typical of planetary imagery.

4.2. Metrics

The metric used during the evaluation and testing included is the common classification metrics call: accuracy.

$$\text{Accuracy} = \frac{\text{Correct predictions}}{\text{All predictions}} \tag{1}$$

As described in Equation 1, accuracy refers to the percentage of correct predictions among all predictions. In order to explain how the accuracy score is calculated, let's consider a region with 4 classes of craters and the associated predictions. We then divide all correct predictions by the total number of predictions.

To have a better insight into the model's intra-class error, we also calculate a confusion matrix. This diagnostic tools provides a detailed estimation of the model's performance by showing the rate of true positives with respect to misclassified identifications. This helps assess misclassification patterns and overall classification reliability. The confusion matrix summarizes the performance of the classifier by comparing predicted and true labels. Each row corresponds to the actual crater class, and each column to the predicted class. Values along the diagonal represent correctly classified samples, while off-diagonal entries indicate misclassifications between classes (e.g., a regular crater predicted as secondary). This structure allows direct visualization of which classes are most frequently confused and forms the basis for computing derived metrics such as accuracy, precision, and recall.

4.3. Training

During the training process, the model was trained from scratch (random initialization of the weight) for 40 epochs, incorporating a learning rate warmup over the initial 5 epochs, followed by a cosine learning rate decay, as shown in Figure 5. This strategy is designed to enhance model convergence and performance.

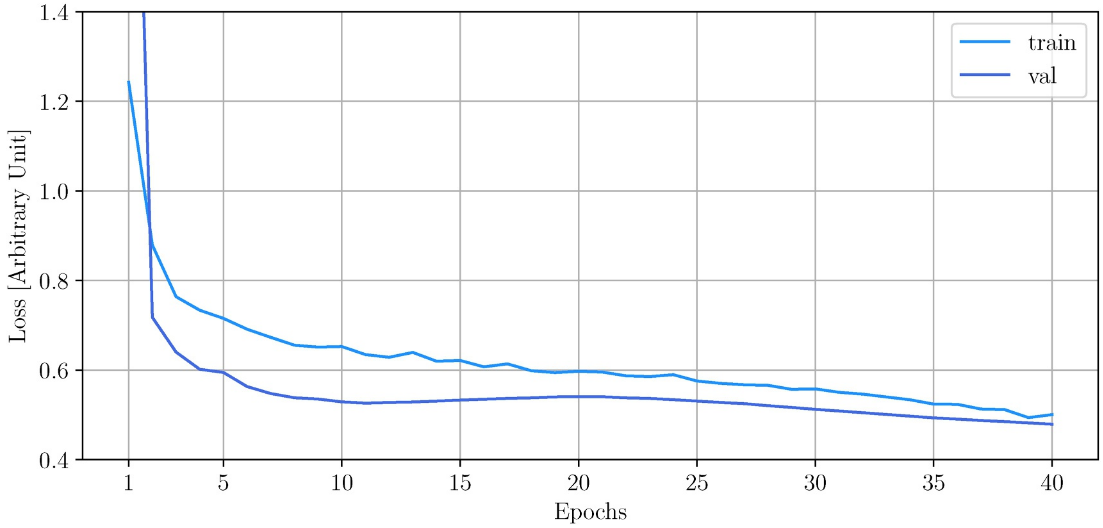

Figure 6. Training and Validation losses with respect to learning epochs.

4.3.1. Learning Rate Warmup

The warmup phase gradually increases the learning rate from a lower starting value to the initial target learning rate over the first 5 epochs. This approach helps stabilize the training process and prevents issues like unstable updates or divergence early in training (Goyal et al., 2018; Kalra & Barkeshli, 2024).

4.3.2. Cosine Learning Rate Decay

After the warmup phase, a cosine annealing schedule is applied to decrease the learning rate smoothly over the remaining epochs. This method reduces the learning rate following a cosine curve, which allows the model to make finer adjustments to the weights as training progresses, potentially leading to better generalization and performance. The cosine decay strategy is known for its effectiveness in achieving smoother transitions in learning rates compared to linear decay methods (Gotmare et al., 2018).

The combination of these techniques—learning rate warmup and cosine decay—result in an optimized training process by initially allowing large updates for rapid learning and gradually refining the model parameters for improved accuracy and stability (see Figure 6).

During the learning phase we set the batch size to 128, to improve the computational efficiency and stabilizing the training process. The batch size value is a crucial hyperparameter. Indeed, from our experience larger batch sizes can lead to more accurate gradient estimates, facilitating smoother convergence and this value is the maximum possible with our GPU architecture.

We also implemented a Multi-Scale Training: The model was trained on images of varying sizes, improving its robustness to objects at different scales. This approach randomly resizes input images during training, helping the model generalize better across diverse object dimensions.

4.4. Validation

At the end of each training epoch, after the model has processed the entire training data set, it evaluates its performance by calculating the top-1 accuracy on the validation data set. This evaluation allows us to monitor the learning progress, as depicted in Figure 7.

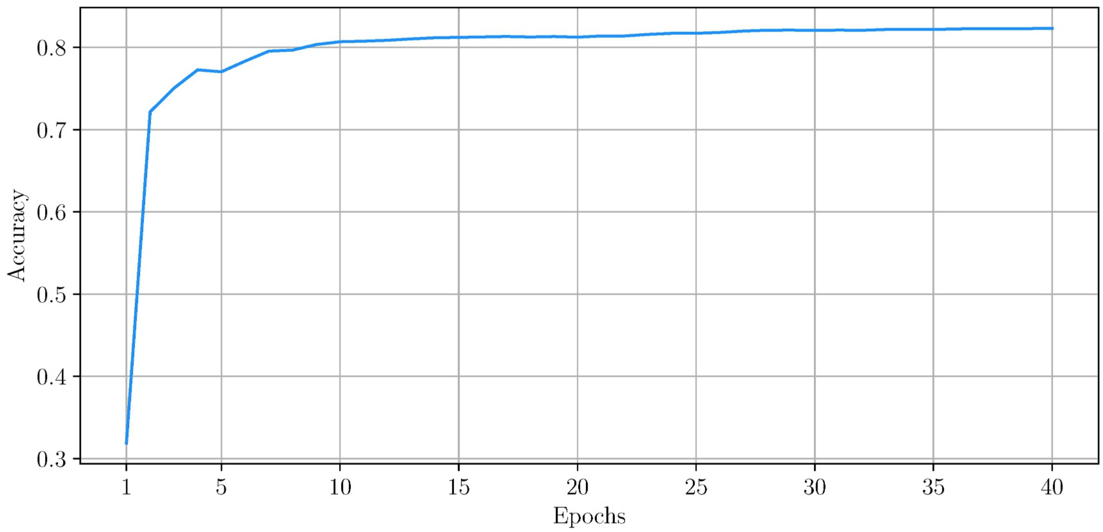

Figure 7. Validation accuracy with respect to learning epochs.

5. Results and Discussion

5.1. Performance

5.1.1. Global Performance

After training, we use our test to evaluate the model on images never seen before. We then measure the confusion matrix, as shown in Figure 8. As illustrate in Equation 1, to evaluate the overall performance of our classifier, we computed the mean accuracy as the ratio of correctly classified instances (i.e., the sum of the diagonal elements of the confusion matrix) to the total number of predictions (i.e., the sum of all elements in the confusion matrix). This gives us a mean accuracy on this data set of 81%.

The distribution of correct and incorrect classifications reflects the trends observed during validation, reinforcing the reliability of our trained model. This consistency suggests that the model has effectively generalized to unseen data, exhibiting minimal overfitting or unexpected deviations. Furthermore, the impact of incorrect classifications on the resulting dating curves is discussed in detail in Section 5.2.

5.1.2. Regional Performance

We conducted an evaluation of our trained model across all quadrangle regions present in the test data set, resulting in the accuracy map shown in Figure 9. This geographical testing is critical as it allows us to assess the spatial consistency and robustness of the classifier across diverse terrains, illumination conditions, and morphological features. By analyzing accuracy at the quadrangle scale, we can detect potential biases due to the latitude.

As shown in Figure 9, some regions present a significantly lower accuracy score compared to others. This pattern is not correlated with the fraction of no data (NaN) (see Figure C1). For instance, in the area between 100° and 140°W longitude and 60° to 80°N latitude, the accuracy drops to around 0.575. In this region, we can count 22 *Ghost*, 118 *Layered* 492 *Regular* and no *Secondary* craters. Many craters were classified as *Ghost* when they actually appear to be *Regular*. Indeed, in Figure 10, on the first line, three out of four craters which were classified as ghost do not clearly exhibit characteristics typical of ghost craters. Their classification as ghost crater is subject to human interpretation. Moreover, the original crater classification database (Lagain et al., 2021), was done manually by more than 60 different human operators. In the light of this, we interpret these quadrangles with a notably low score as due to “outliers” in human classification.

We can also see in Figure 9, that certain regions exhibit a notably higher classification accuracy, such as the quadrangle defined by longitudes 156° to 164°E and latitudes 16° to 24°S, where the accuracy reaches 0.808 (see Figure 11). In this quadrangle, Lagain and his collaborators identified 355 *Regular* craters, 12 *Layered* craters, 5

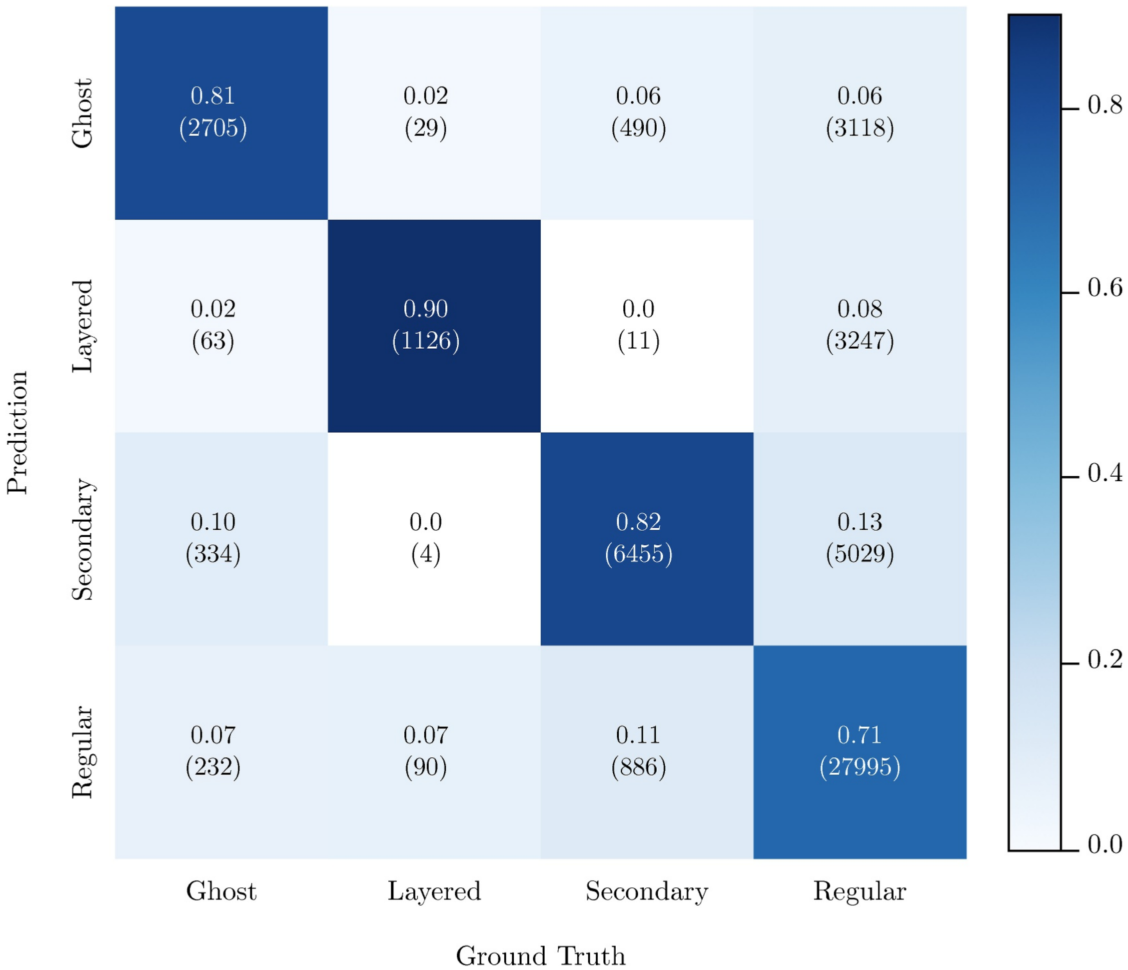

Figure 8. Confusion matrix made on the test subdataset. These results show for instance that 81% of true Ghost crater–2705 craters—were correctly classified, and 2% of them (64 craters) where misclassified as Layered. Overall, the performance is excellent (81%). The regular craters appear slightly more difficult to classify, maybe due to human misclassification (see Figures 10 and 11).

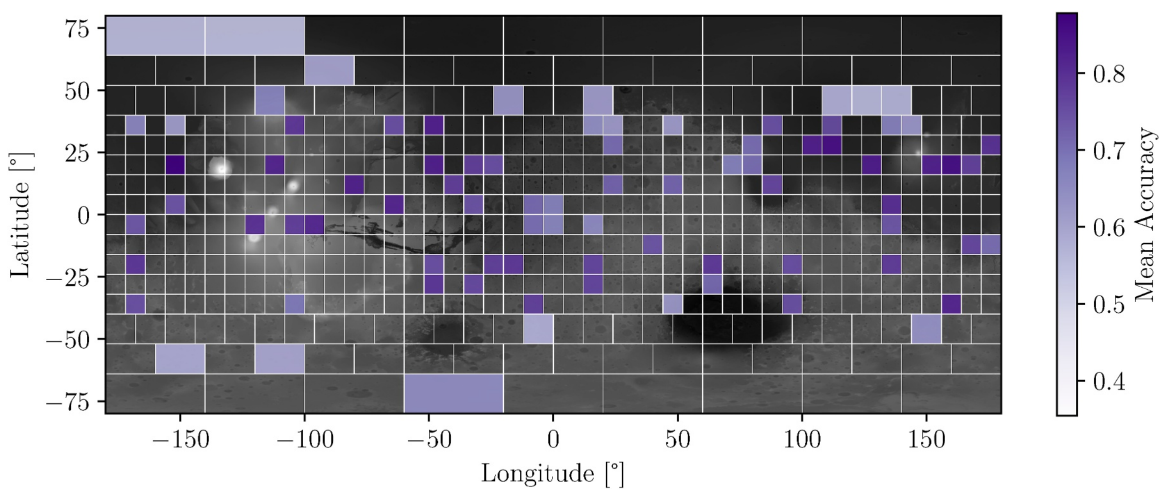

Figure 9. Global accuracy on each quadrangle of the test database which contains 51,814 craters as described in Table 2.

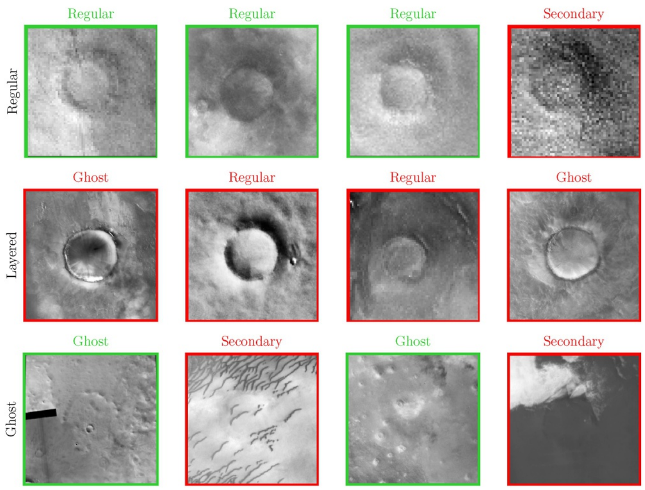

Figure 10. Illustration of the lowest accuracy case. Example of 12 craters present in a test area between 100° and 140°W longitude and 60° to 80°N latitude, which were, from top to bottom, classified as Regular, Layered and Ghost in the reference data set. The title on top corresponds to the prediction made by our classification algorithm and the color offers a visual representation of whether the classification is right or wrong with respect to the reference database.

Ghost craters, and 4 *Secondary* craters. The exceptional performances observed here is largely attributed to the predominance of *Regular* craters, which constitute 94.4% of the craters in this region. Furthermore, among the predictions made for this crater class, 84.5% were correctly identified as *Regular* craters.

While a manually classified global catalog (Lagain et al., 2021) already exists, its resolution is limited to craters larger than ∼1 km and was generated through time-consuming visual inspection. The present work provides a complementary, automated solution capable of classifying approximately 385 craters per second with our configuration described in Section 3.3, including those smaller than 1 km, with consistent accuracy and repeatability. This new capability greatly accelerates crater-based geological analyses and extends the accessible size range for age estimation, thereby enhancing our understanding of surface evolution on Mars and other planetary bodies.

5.1.3. Sources of Apparent Errors and Regional Variability

Beyond isolated labeling inconsistencies, a substantial fraction of the apparent "misclassification" reflects geomorphological under-determination: for degraded rims, layered ejecta, or clustered small craters, multiple labels can be reasonably defended by experts, which makes the decision boundary inherently fuzzy. In our experiments we did not observe a consistent, data set-wide penalty attributable to class imbalance, image quality, shadowing, or topography once the stratified split and standard augmentations were applied. Accuracy drops in the lowest-performing quadrangles are not driven by a single cause, but rather by local combinations of resurfacing history and low-contrast textures that increase ambiguity.

One clear failure issue, however, is the presence of missing-data areas in the CTX mosaics (pixels flagged as no-data), which can induce spurious assignments; this behavior is consistent with our robustness tests and with the explicit augmentation that injects black patches. Altogether, these effects confirm that the classifier can be regarded as a robust quick and first-level filtering tool, while manual validation remains advisable for high-precision dating studies, particularly in geologically complex regions.

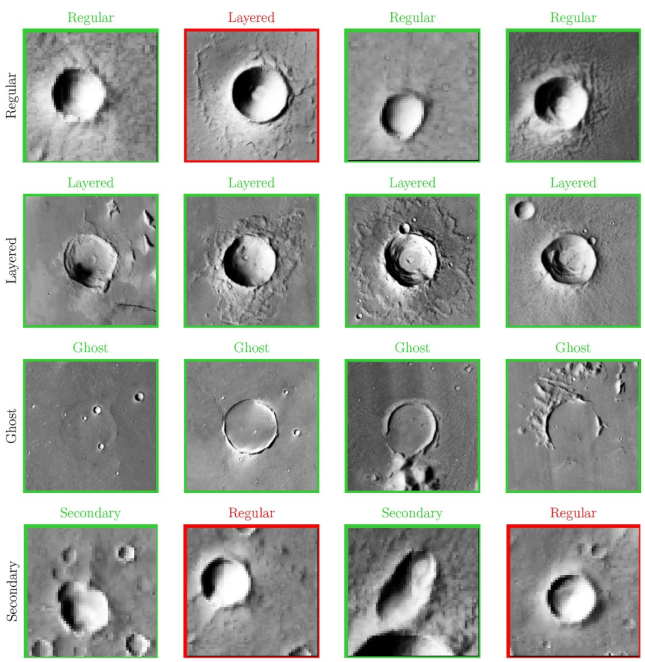

Figure 11. Illustration of the highest accuracy case. Example of 16 craters present in a test area between 156° and 164°E longitude and 16° to 24°N latitude, which were, from top to bottom, classified as Regular, Layered, Secondary and Ghost.

5.1.4. Inference at Smaller Scale

Since impact processes follow scaling laws (Melosh, 2011), we expect to have similar classification performance at scales lower than the training data set. As expected, the performances of our algorithm do not depend on crater diameter in the test data set (see Table 3). Still, at metric scales, erosion may be more efficient and degrade the performance. This was not evaluated in this work.

The detection limit indeed depends more on the number of pixels rather than the actual pixel size of the crater. Based on detection algorithm (Martinez et al., 2025b), we could infer that this limit is 10 pixels/crater. Here we limit the analysis on crater diameter ≥500 m, on image with 50 m pixel resolution, reaching thus the limit of 10 pixel/crater.

A smaller scale classification database would be very useful to evaluate precisely this point. The building of such a database would be an interesting future work perspective.

5.2. Application: Dating and Analysis of Utopia Planitia

The Zhurong rover, part of the Tianwen-1 mission, landed on 14th May 2021 in southern Utopia Planitia (25.066°N, 109.925°E) within the Amazonian

Table 3
Mean Accuracy With Respect to the Minimum Crater Size of the Crater Database (in Diameter)

	ø > 1 km	ø > 10 km	ø > 100 km	ø > 300 km	ø > 500 km
Mean accuracy	0.81	0.8125	0.8125	0.82	0.79

Note. We can see that our classifier performance is robust with respect to the crater size.

lowlands of Mars. This region is characterized by relatively smooth terrains, interpreted as resurfaced plains, with a sparse population of large craters but abundant small features such as secondary craters and pitted cones. Previous studies based on cumulative crater size–frequency distribution (CSFD) analysis have suggested resurfacing ages ranging from early to middle Amazonian, with significant variability depending on the chosen count area and interpretation of crater morphologies (Tanaka et al., 2014).

In our analysis, we applied a deep-learning–based detection (Martinez et al., 2025b) followed by our geomorphological classification pipeline (this study). We restricted the count area to the CTX quadrangle presented in Figure 12, focusing on craters with diameters above 100 m to ensure robust detection and minimize classification uncertainties. In 4, we can see the distribution of the 146,041 craters detected and classified in the area.

Since impact processes are scaling (Melosh, 2011), we expect to have similar classification performance at scale lower than the training data set. The resulting inference map is presented in Figure 12. The CSFD curve shown in Figure 13, was fitted using the (Hartmann, 2005) chronology model to estimate the model surface age.

Our CSFD yields to an over-estimation in comparison to manual counting when using all detected craters (see Figure 13a), but also the crater density is larger than expected theoretically in the range 2–10 km. When filtering out *Secondary* and *Ghost* craters, this effect is significantly decreased and the density curve better follow the theoretical curve (see Figure 13b).

The absolute model age $\sim 2.92^{+0.11}_{-0.13}$ Gy (Early Amazonian) we estimate is consistent with the lower end of the range proposed by Wu et al. (2022), who reported ages $\sim 3.01^{+0.23}_{-0.47}$ y on a manually crater counted analysis. Notably, our results reinforce the interpretation that the plains in southern Utopia Planitia underwent significant resurfacing during the Amazonian (around ~ 2.12 Gy), most likely through processes such as mud volcanism, ice-related sediment mobilization and volcanic flooding, which effectively reset the small-crater population while preserving larger, older impact structures.

Figure B2 show the age determination of the Lagain et al. (2021) database on the very same surface unit of Utopia Planitia. The left plot shows all crater and the right one only considered the crater classified by our tool as regular and layered. The right plot shows a better agreement with the expected theoretical model, especially in the 2–10 km diameter range. In addition, the CSFD and the age (3.28 Gy) are larger than manual analysis from Wu et al. (2022).

5.2.1. Effect of Misclassification on Age Estimation

Although the classifier achieves an overall accuracy between 71% and 90% (Figure 8) depending on the classes, the confusion matrix reveals class-dependent misclassifications that may influence crater count–based age estimations. To address concerns regarding the potential impact of classifier errors on crater-based surface dating, we conducted a series of controlled experiments to quantify how misclassification of ghost and secondary craters influences age estimation. As shown in Figure 8, our classifier exhibits a 10%–29% error rate across crater classes, which translates to two primary types of errors: (a) the incorrect removal of inferred ghost or secondary craters (false negatives for regular/layered classes used for age estimation) or (b) the incorrect retention of inferred regular or layered craters (false positives for regular/layered classes used for age estimation).

Based on the confusion matrix for our test set, we observed that 45% of inferred ghost/secondary craters were incorrectly classified, because labeled as layered or regular in the database:

Table 4
Total Number of Craters in the Zhurong Landing Area, Classified Using Our YOLO-Based Approach

	Number per class
Regular	30,252
Layered	308
Secondary	114,438
Ghost	1,063
Total	146,061

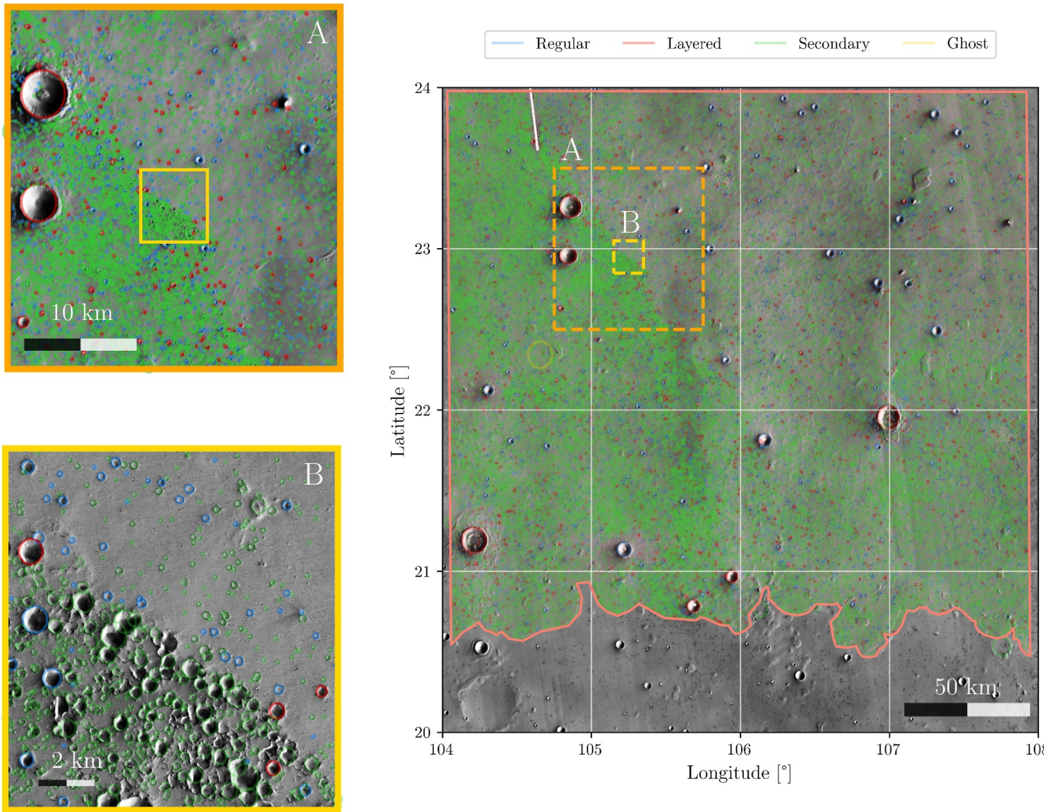

Figure 12. Detection and classification on a 4° × 4° quadrangle in Utopia Planitia (equirectangular projection; lower-right coordinate: 20°N, 104°E). All craters were detected using the method described in Martinez et al. (2025b). The classification has been performed with the tool presented in this paper. Blue and red circles denote *Regular* and *Layered* craters, respectively, as classified by our algorithm; these craters are retained for surface dating. Green and yellow circles represent *Secondary* and *Ghost* craters, respectively, which are excluded from the dating analysis. Zones A and B are magnified areas for detailed examination.

$$\frac{29 + 3,118 + 4 + 5,029}{2,705 + 29 + 490 + 3,118 + 334 + 4 + 6,455 + 5,029} = \frac{8,180}{18,164} = 0.450$$

while 3.5% of inferred regular/layered were incorrectly classified, because belonging to the ghost/secondary classes in the database:

$$\frac{63 + 11 + 232 + 886}{63 + 1,126 + 11 + 3,247 + 232 + 90 + 886 + 27,995} = \frac{1,192}{33,650} = 0.035$$

These misclassifications could, in principle, introduce biases in cumulative crater size frequency distribution (CSFD) analyses, thereby affecting age estimates. To empirically assess this impact, we performed crater detection and classification on a selected region as presented in Section 5.2 and filtered the data set to retain only craters classified as Regular or Layered to estimate the CFSD and age. We then compared the resulting age estimate with manual counts reported by Wu et al. (2022). To further investigate the sensitivity of our age estimates to classification errors, we designed three experiments simulating the worst-case scenarios of misclassification:

- **Experiment 1** (Bias on Ghost/Secondary Craters): To simulate the scenario in which a fraction of inferred ghost and secondary craters are erroneously removed during the age estimation process, we randomly add a fraction of them artificially in the age estimation pool.

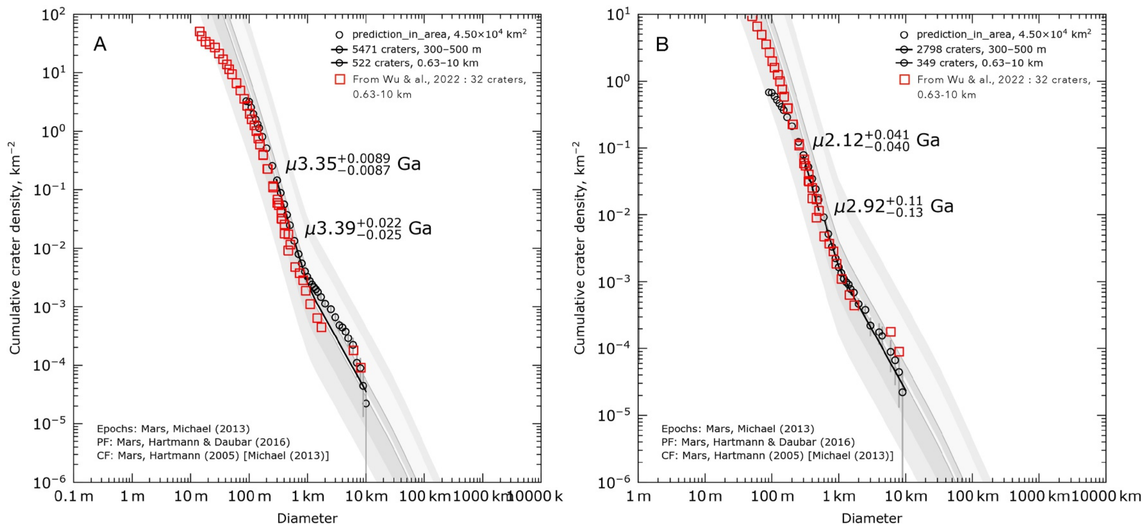

Figure 13. (a) Cumulative crater size frequency distribution (CSFD) and age determination derived from our crater detection algorithm (CDA), as presented in Martinez et al. (2025b), compared with the results of Wu et al. (2022) (red dots) for a neighboring region near the Zhurong landing site. For crater diameters ranging from 2 to 10 km, the data do not conform to the model age, and the observed density is systematically higher than that reported by Wu et al. (2022). (b) In this analysis, we applied our crater classification algorithm to filter the detection obtained using our CDA, specifically removing *ghost* and *secondary* craters. Only craters classified as *regular* or *layered* were retained. The resulting crater density aligns closely with the findings of Wu et al. (2022) (red dots), and the previously observed over-density in the 2–10 km diameter range is eliminated. As commonly refereed in the crater counting application, "CF" and "PF" refer respectively to the chronology function and crater production function used in the craterstats algorithm Hartmann (2005); Michael (2013). "Epoch" refers to the agreed Martian epochs chronology and is unused in these graphs.

We randomly selected 49.6% of inferred ghost craters—that is, the prediction was *ghost* but the ground truth was either *regular* or *layered*—and count them in the age estimation. This fraction was calculated as:

$$\frac{29 + 3118}{2705 + 29 + 490 + 3118} = 0.496$$

Similarly, we randomly selected 42.6% of secondary craters—that is, the prediction was *secondary* but the ground truth was either *regular* or *layered*—and count them in the age estimation. This fraction was calculated as:

$$\frac{4 + 5029}{334 + 4 + 6455 + 5029} = 0.426$$

These final filtered data set will thus be larger, with more craters and an older age.

- **Experiment 2** (Bias of Regular/Layered Craters): To simulate the scenario in which a fraction of inferred regular and layered craters are erroneously included from the age estimation process we randomly removed a fraction of them artificially in the age estimation pool.

We randomly selected 3.8% of inferred regular crater—that is, the prediction was *regular* but the ground truth was either *ghost* or *secondary*—and discard them in the age estimation. This fraction was calculated as:

$$\frac{232 + 886}{232 + 90 + 886 + 27995} = 0.038$$

Similarly, we randomly selected 1.7% of inferred layered crater—that is, the prediction was *layered* but the ground truth was either *ghost* or *secondary*—and discard them in the age estimation. This fraction was calculated as:

$$\frac{63 + 11}{63 + 1126 + 11 + 3247} = 0.017$$

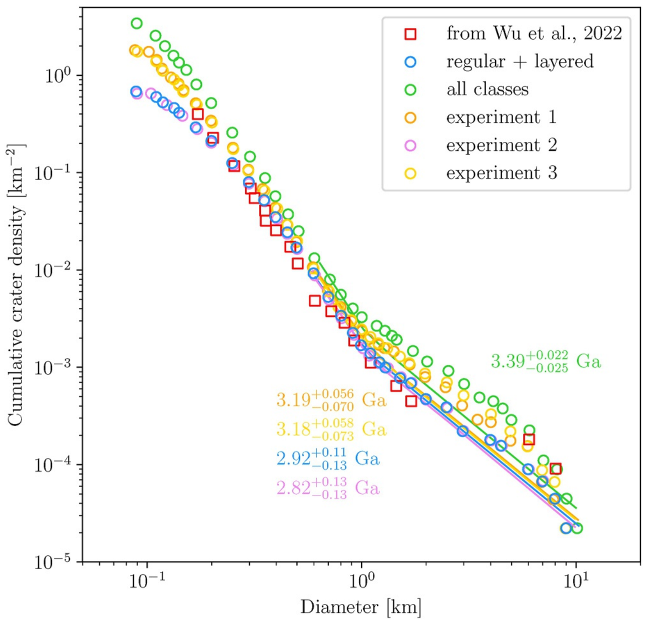

Figure 14. Cumulative crater size-frequency distributions (CSFDs) for different crater data sets and experimental conditions near the Zhurong landing site. The plot compares the cumulative crater density as a function of crater diameter for multiple experiences describe in Section 5.2. The fitted isochrons and their corresponding absolute model ages (Ga) are displayed for each data set, with uncertainties indicated. The results highlight the sensitivity of age estimates to the inclusion or exclusion of specific crater classes.

These final filtered data set will thus be smaller, with less crater and an younger age.

- **Experiment 3** (Combined bias): We combined the conditions of Experiments 1 and 2, simultaneously adding misclassified ghost/secondary craters and removing misclassified regular/layered craters from the filtered data set. This represents a more comprehensive evaluation of how cumulative classification errors might affect age determination.

The results of these experiments, illustrated in Figure 14, demonstrate that Experiment 1 produces an age 3.19 Gy, Experiment 2: 2.82 Gy and Experiment 3: 3.18 Gy to be compared with our formal age determination of $2.92^{+0.11}_{-0.13}$ Gy.

The bias in estimated surface ages remains small (+0.27 for Experiment 1, −0.10 for Experiment 2, −0.26 for Experiment 3) and around 2 times the formal uncertainties $+0.11_{-0.13}$.

Despite the introduction of simulated misclassification errors, the derived ages show minimal deviation suggesting that while classifier errors do introduce a lot of small craters misclassification, their impact on crater-based chronology is limited, particularly when considering the broader geological context and the inherent uncertainties in CSFD analyses.

These findings underscore the resilience of our automated classification pipeline for geological applications, even in the presence of non-negligible misclassification rates. We shows that the formal age uncertainties can be larger than the one from usual curve fitting (Michael, 2013) due to misclassification of crater, with an total uncertainties augmented by a factor of ∼2.

This limitation could be surpassed by further classification refinements—such as targeted improvements in classifier accuracy for smaller craters or the integration of additional geomorphological constraints—to enhance the precision of future age estimations.

6. Conclusion

In this study, we presented an innovative deep-learning approach using the YOLOv11 architecture for automatic crater classification on Mars. Our methodology use high-resolution imagery from the Mars Reconnaissance Orbiter's CTX camera and a refined crater database. The classification targeted four geomorphological crater categories defined by Lagain et al. (2021): *regular*, *ghost*, *layered*, and *secondary*. A notable pre-processing step using Hough Circle detection significantly enhanced the alignment between the ground truth and observed crater features, improving the quality and reliability of the classification model.

The proposed YOLOv11-based model achieved a robust overall accuracy of 81%, demonstrating its capability to reliably classify various crater types on a planetary scale. We observed, however, geographic variations in accuracy, particularly noticeable in the latitudes ranges 60° to 80°N and longitudes 100° to 140°W, highlighting areas requiring further refinement and analysis.

Our methodology sets a precedent for fully automated planetary surface analyzes, significantly reduces human subjectivity and saves time compared to manual annotation techniques traditionally used in planetary geomorphology. This automated classification not only accelerates the process but also provides consistent and reproducible results that can be systematically applied to large data sets.

An a real example on the Utopia Planitia (near the Zhurong landing site), we found that removing automatically the ghost and secondary crater lead to better fit to the expected CSFD model. The effect of misclassification of crater type that must be taken into account (regular and layered) and should be discarded (ghost and secondary) from the age estimation is relatively reduced. On the same example, we show that the bias is with ∼2 times the formal uncertainties from usual cratering model (Michael, 2013).

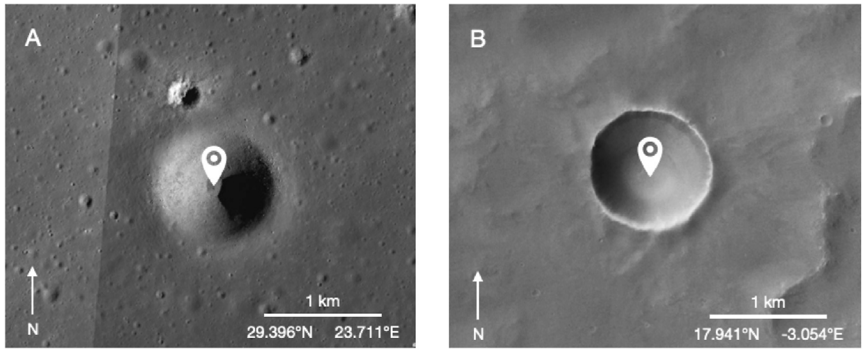

Figure 15. Comparison between two 1 km size crater on the Moon (a) and on Mars (b).

To go further with the capabilities of this classification tool, several ideas might be developed in the future. One possibility would be to apply our model to other planetary bodies, such as lunar craters. Due to significant geomorphological differences between the lunar and martian surfaces, applying the model to the Moon may require additional targeted training, particularly for small craters. As illustrated in Figure 15, *regular* lunar craters display different morphological characteristics compared to those on Mars. The absence of atmosphere and the effects of space weathering on the Moon result in much smoother, more eroded and diffuse crater rims. The typical bowl-shaped morphology is often less defined, and a specific configuration should be developed, either by training with supplementary data, or by creating a new class.

Similarly, exploring the model's applicability to other planetary bodies such as Mercury or icy moons like Europa and Enceladus would further demonstrate the robustness and versatility of our deep-learning pipeline. Such expansion may require the usage of customized data sets that we will only have after the first light from the Europa Clipper and JUICE missions. Unfortunately, no existing ground truth database are available to assess the performance of our model.

Additionally, the use of transfer learning techniques could substantially help to adapt our model to different crater databases and imaging sources.

Lastly, an exciting potential future application of our methodology lies in refining planetary surface age determination techniques. By accurately identifying and excluding secondary and ghost craters, which tend to bias crater density measurements, our automated classification approach could lead to more precise age determinations, thereby substantially advancing our understanding of planetary geological chronologies.

In conclusion, our deep-learning-based automatic crater classification algorithm represents a major advancement in planetary geosciences, providing a powerful tool to systematically characterize planetary surfaces. The continued refinement and expansion of this approach hold considerable promise for improving our understanding of the geological history and evolution of Mars and other planetary bodies.

Appendix A: Circle Detection Based on Hough Transform to Improve Crater Location and Radius

One of the defining features of craters is their circular shape. To improve the accuracy of our crater database, we propose implementing an automated circle detection method to estimate the best-fitting crater circle near the initial ground truth. This involves first generating an edge map of the image, followed by applying a Hough transform to the image edges. The Hough transform is an elegant technique that requires only a few parameters to identify circles in an image. In this section, we describe our approach to boundary detection, aimed at refining crater positions and radii.

A1. Edge Detection

The first step in our crater refinement pipeline is to generate an edge map of the image. This binary map highlights potential crater boundaries by identifying non-zero pixels as edge points. We use the Canny edge detector from

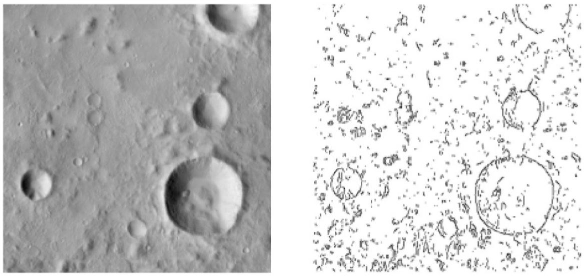

Figure A1. Canny edge filtering. On the left we can see a CTX image of the surface of Mars. On the right the binary image produced by the Canny edge filtering. Each black point is considered as a potential crater circle boundary.

the Python scikit-image library to perform this task (Pedregosa et al., 2011). This method calculates gradient intensity via a Gaussian derivative filter. Before applying the edge detector, we crop the image around the crater of interest and resample it according to its diameter. This step helps standardize scale and reduce noise. We noticed that large craters (radius >80 pixels) often contain many unrelated edges, which can mislead the detection algorithm.

To improve contrast, we normalize image brightness by stretching pixel values to have a mean of 128. We adjust the standard deviation of this stretch depending on the crater type. For regular craters, we set the standard deviation to 9. For heavily eroded "ghost" craters, we increase it to 15 to enhance their visibility. These ghost craters tend to have faint rims and are harder to detect. The Canny filter includes two thresholds: a low and a high value. Pixels above the high threshold are marked as edges, and those below the low threshold are discarded. Intermediate pixels are only considered edges if they connect to strong edges. Threshold values are set to (30, 45) for ghost craters and (35, 40) for others. An example of a Martian surface image and its corresponding edge map is shown in Figure A1.

A2. Hough Transform

The next step is to estimate all possible circles present in the binary edge map. Detecting circles from images such as Figure A1 presents three main challenges. First, we must determine which edge points actually belong to the crater we aim to fit. Second, we often encounter incomplete data—either due to missing or corrupted parts of the image, or due to imperfect edge detection. The latter can occur in regions with little or no gradient variation, where edges fail to be properly identified. Finally, we must account for noise, which can introduce spurious edges into the map.

The Hough transform offers a robust and unified solution to these challenges (Duda & Hart, 1972; Hough, 1962). It is a powerful technique for detecting simple geometric shapes, such as straight lines and circles, even in noisy or incomplete data. The key idea is to map each point in the image space to a corresponding curve in a parameter space that represents potential circles. Peaks in this parameter space indicate the presence of circular features in the original image.

Indeed a circle, that is all points (x, y) in the circle, can be described by the following equation:

$$(x - x_h)^2 + (y - y_h)^2 = r_h^2 \tag{A1}$$

with:

- (x_h, y_h) as the coordinate of the center of the circle that we want to fit from edge map
- r_h as the radius.

Subscript $_h$ denotes the crater after Hough transform (updated circle) and $_0$ denotes the initial position.

The Hough transform is used to estimate the optimal circle parameters—center coordinates $(x_h,\ y_h)$ and radius r—that best match the edge pixels in the image, thereby helping to accurately locate and size craters. As illustrated in Figure A2, the algorithm transforms each edge point in image space into a corresponding set of possible circles in a three-dimensional parameter space.

This parameter space is represented by an accumulator array, Acc, indexed by (x_h, y_h, r). Each entry in this array counts the number of "votes" for a specific circle defined by its center and radius. For a given radius, the algorithm iterates over all edge pixels and casts votes for all possible circle centers (x_h, y_h) that could correspond to that pixel. This process is repeated across a range of radii, so that circles supported by many edge points accumulate more votes.

The circles with the highest number of votes in the accumulator array are considered the most likely to correspond to actual crater rims in the image. Once all edge pixels have been processed, the best-fitting circles are those

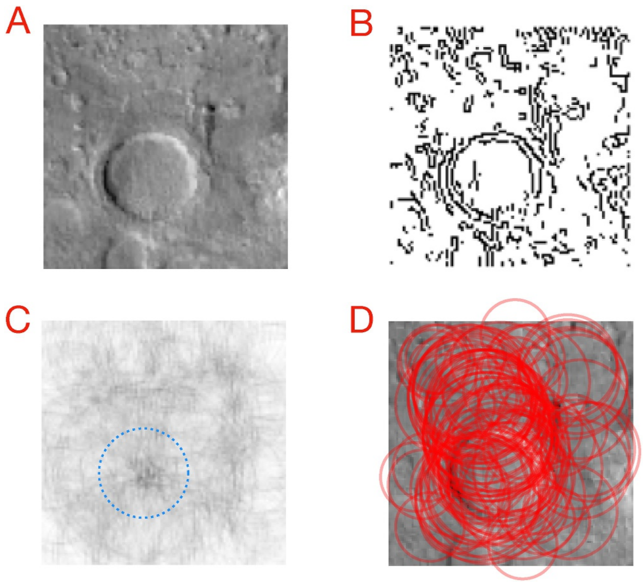

Figure A2. Hough circle detection perform on crater n° 06–002993 from quadrangle E108/N40. Main Hough transform process steps. (a) reference image (b) edge-map; (c) accumulation in gray level (darker for higher accumulation) of the center location at a given radius, indicated by the blue circle. (d) Plots of the 100 best accumulations, that is the 100 best circles in the initial image space, each one with an accumulation score.

associated with the highest values in the accumulator, *Acc*. This array can be interpreted as a scoring space: the higher the score, the better the corresponding circle aligns with the crater edges in the binary map. The main steps of the Hough transform process are illustrated in Figure A2.

As shown in Figure A2d, the Hough circle algorithm tends to detect a large number of crater candidates, even within a relatively small image. To ensure accurate detection and reduce ambiguity, we apply the algorithm to small image patches centered on each crater listed in the database. Running the Hough transform on large planetary regions would result in many edge detections and, consequently, a high number of potential crater candidates. This not only increases computation time significantly but also risks confusing neighboring craters as alternative matches. To address this, we limit the analysis to localized patches around known craters, as illustrated in Figure A5.

A3. Best Circle Selection

As discussed in Appendix A2, the Hough transform does not return a single solution, but rather a set of candidate circles that each fit the crater to varying degrees. Since our goal is to make only a small correction to the values in the existing database, we want our final selection to remain close to the original crater parameters. In other words, we trust the database but aim to slightly refine the crater's position and size. At the same time, we want to select the candidate with the highest geometric consistency—i.e., the best circularity. The objective is therefore to identify the optimal circle among the candidates.

To do this, we define a new scoring function that combines multiple criteria as illustrate in Figure A3:

1. The candidate center should be close to the reference center.
2. The candidate radius should be close to the reference radius.
3. The candidate circle should have the highest Hough accumulation score.

We express these criteria as probabilities and combine them to select the most likely match.

A3.1. Center Distance Probability

To favor candidates located near the reference center, we introduce a distance-based factor that can be interpreted as a pseudo-probability, normalized to a maximum value of 1. This factor is based on the Euclidean distance Δ_d between the reference center and the candidate center. We model this using a Gaussian-like function, where the standard deviation σ_d represents the acceptable error range around the reference center. An illustration of this function is shown in Figure A4.

$$P_d(\Delta_d, \sigma_d) \ = \ \exp\left(-\left(\frac{\Delta_d}{\sigma_d}\right)^2\right) \tag{A2}$$

We used defined the local acceptance on the center, using the initial radius with:

- Δ_d: the list of distance between each candidate center and reference center as shown in Equation A3.
- σ_d: the characteristic distance deviation as describe in Equation A4

$$\Delta_d = \sqrt{(x_h - x_0)^2 + (y_h - y_0)^2} \tag{A3}$$

$$\sigma_d = \begin{cases} 40 & \text{if } r_0 \ \leq \ 20 \text{ pixel} \\ 2 \times r_0 & \text{otherwise} \end{cases} \tag{A4}$$

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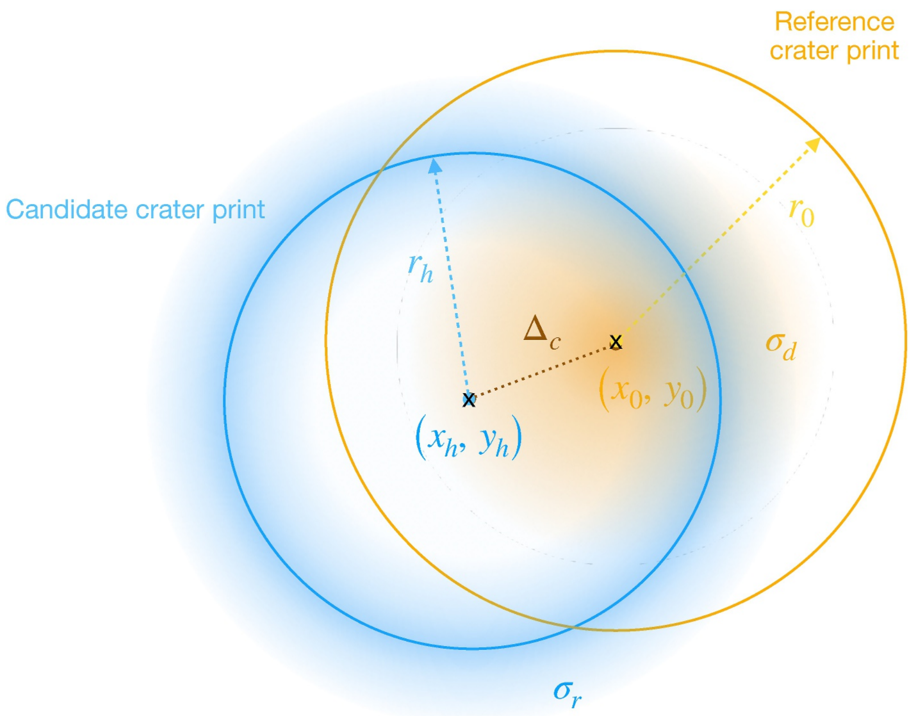

Figure A3. Schematic view of the problem. In blue we have the candidate crater with its center position (x_h, y_h) and its radius r_h. In orange we have the reference crater with its center position (x_0, y_0) and its radius r_0.

In this way, $P_d(\Delta_d, \sigma_d)$ will give less weight to the center candidates furthest from the reference center during the selection process.

A3.2. Radius Distance Probability

To favor candidates whose radius is close to the reference value, we introduce a radius-based factor that can be interpreted as a pseudo-probability, normalized to a maximum value of 1. This factor is based on the absolute difference Δ_r between the reference radius and the candidate radius. As with the center distance, we model this

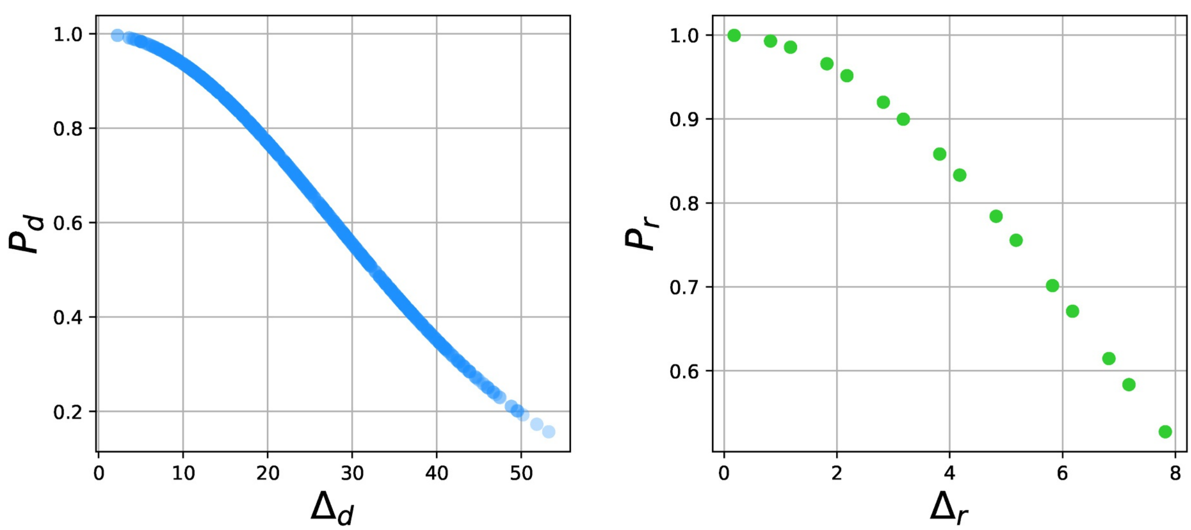

Figure A4. Left: Center distance probability; Right: Radius distance probability.

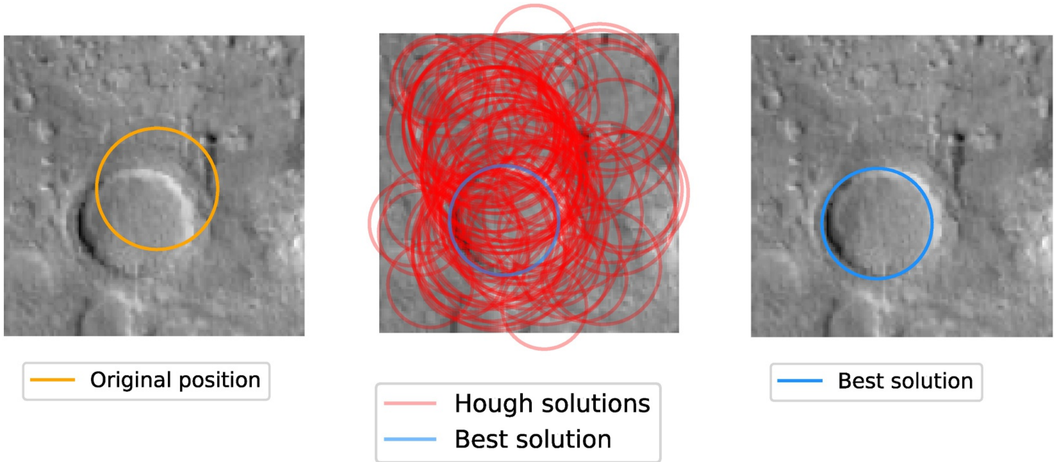

Figure A5. Hough circle detection perform on crater n° 06–002993 from quadrangle E108/N40. Left: Crater Initial crater. Middle: We plot the 100 craters from the Hough transform. Right: best candidate crater, using Equation A9.

using a Gaussian-like function, where the standard deviation σ_r defines the acceptable range of deviation around the reference radius. The shape of this function is also illustrated in Figure A4.

$$P_r(\Delta_r, \sigma_r) = \exp\left(-\left(\frac{\Delta_r}{\sigma_r}\right)^2\right) \tag{A5}$$

We defined the local acceptance on the center, using the initial radius with:

- Δ_r: the list of differences between each candidate radius and reference radius as shown in Equation A6.
- σ_r: the characteristic distance deviation as describe in Equation A4

$$\Delta_r = |r_0 - r_h| \tag{A6}$$

$$\sigma_d = \begin{cases} 15 & \text{if } r_0 \le 7.5 \text{ pixel} \\ \frac{r_0}{2} & \text{otherwise} \end{cases} \tag{A7}$$

In this way, the function $P_r(\Delta_r, \sigma_r)$ assigns lower weights to candidate radii that deviate more from the reference radius during the selection process.

A3.3. Circularity Score

As a third factor, we introduce a term that reflects the quality of the circle detection itself. This factor is derived from the Hough accumulator score, *Acc*, which can also be interpreted as a pseudo-probability normalized to a maximum value of 1. It favors candidates that are better supported by edge evidence in the image.

$$P_c = Acc \tag{A8}$$

The higher this score will be, the better the candidate circle will fit a circular shape in the edge map.

A3.4. Best Score

We use a combination of this three pseudo-probability by the product of them:

$$P = P_d \cdot P_r \cdot P_c \tag{A9}$$

In this way, the final probability score represents a compromise between three objectives: remaining close to the original crater position, having a radius similar to the reference radius, and achieving a high Hough accumulation

Table A1
Localization of the Quadrangles 4° × 4° That We Used in Order to Verify That Our Circle Detection Algorithm Work

Longitude	Latitude	Region	Number of crater
Verification data set information			
108–112°W	40–44°N	Alba Patera	54
168–172°W	48–52°S	Arcadia Planitia	120
20–24°E	44–48°S	Noachis Terra	139
40–44°E	12–16°N	Noachis Terra	196
96–100°E	60–64°N	Vasistas Borealis	39
108–112°E	40–44°N	Utopia Planitia	66
144–148°E	44–48°N	Utopia Planitia	25

score. The selected circle is the one that maximizes this combined score. If no image is available at the location of a given crater, we simply skip the correction process for that crater. This is not considered a limitation, as such craters will not appear in the training data set and thus have no impact on the model's performance.

A3.5. Limitations

In rare cases, the circle detection algorithm fails simply because no image data is available at the crater's location. This occurs in areas of the CTX data set where data is missing or incomplete. In such cases, we default to the original reference position and radius values without applying any correction. These craters are excluded from further analysis and will not appear in the training data set, so this does not affect the overall performance of the method.

A3.6. Connection to Crater Classification

The refined circle detection described above plays a key role in improving the accuracy and consistency of the crater database. By slightly adjusting the crater center and radius based on image content, we ensure that each crater is optimally localized. This correction directly impacts the quality of the bounding boxes used to extract image patches around each crater. These bounding boxes, defined from the final circle parameters, are then used to generate input samples for the classification model. Ensuring that the crater is properly centered and scaled within the patch improves the model's ability to learn meaningful morphological features. This step is therefore crucial in bridging geometric detection and machine learning, enabling a cleaner and more reliable training data set for crater classification.

A3.7. Evaluation

In order to verify the accuracy of our crater database algorithm, we perform an exhaustive visual verification. We consider all craters from 7 different quadrangles, representative of Mars, ranging from all latitudes and including terrains of different ages, from Noachian to Amazonian. This corresponds to 639 craters as shown in Table A1. Among this 639 craters, 479 were qualified as "Valid" craters by Lagain et al. (2021), 101 as "secondary," 36 as "ghost" craters (i.e..,: eroded craters) and 23 as "layered" craters as summarized in Table A2. We can see in this table that we took quadrangles from various region of Mars, at various latitude/longitude/surface age. We can also see in Figure A6 that our sample have a characteristic crater size-frequency distribution which make it representative of the crater population of Mars.

We divided the result into four categories:

- **Correct, improved**: It means that the final crater is correct and it has been improved by the algorithm.
- **Correct, no change**: It means that the final crater is correct but was not improved since the initial reference position and radius were already correct.
- **Incorrect, degraded**: It means that the circles detected in the image are not correct craters and so it will degrade the database.
- **Algorithm failed, no data**: It means that the algorithm failed due to the lack of data. It could happen when there is a significant area filled with no data value. Also, for a crater position near the bounds of the quadrangles, it could be when the crater is located too far outside the quadrangle. In this case, we leave values from Lagain et al. (2021) untouched.

Table A2
Crater Count by Type in Our Verification Sample

Type	Count
Valid	479
Secondary	101
Ghost	36
Layered	23
Total	639

Our results are presented in Table A3 and Figure A7 suggest that when data is available, the algorithm proposes correct positions and radii for a large majority (96.8%) of the craters. The crater description is improved for 87.6% of cases. In some case, the algorithm failed due to no data (3.8% of the total amount of craters).

As we can see in Figure A7, there is no particular crater size in which our algorithm failed more since it follows the initial crater-size distribution (Figure A6).

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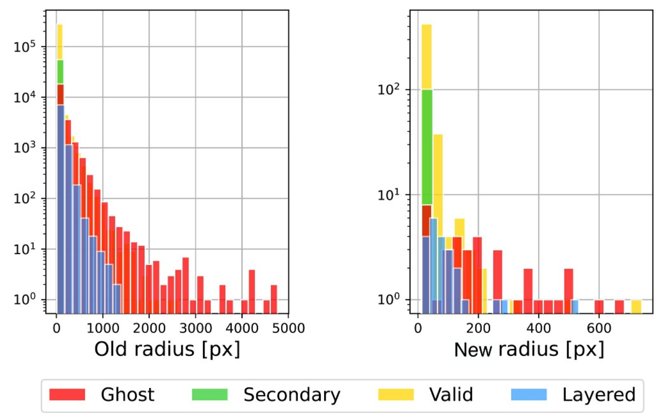

Figure A6. Left: Distribution of the craters with respect to their radius and their crater status as define by Lagain et al. (2021) for the whole planet; Right: Same distribution but for our sample. "Old radius" abscise means that we did make this distribution on the radius define by Lagain et al. (2021).

Table A3
Results for Each Category

Correct, improved	539	84.3%	87.6%
Correct, no change	56	8.8%	9.1%
Incorrect, degraded	20	3.1%	3.3%
Algorithm failed, no data	24	3.8%	
Total	639	100%	100%

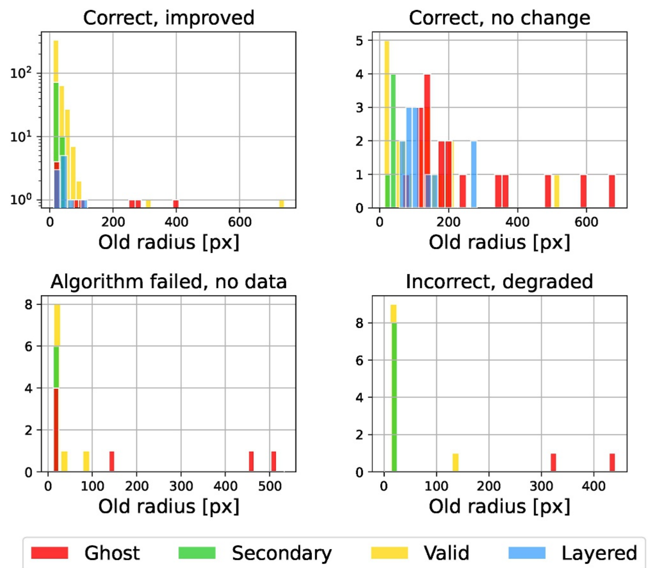

Figure A7. Histograms showing the result of our manually checked verification on 639 craters. The "**Correct, improved**" category put together craters in which the algorithm improve the database. The "**Correct, no change**" category put together crater on which the algorithm worked but which not improve noteworthy the database. The "**Incorrect, degraded**" category put together crater on which the proposed crater is incorrect and degrades the database. The "**Algorithm failed, no data**" category put together crater on which the algorithm failed because of no data. We discriminate each crater category define by Lagain et al. (2021).

In Figure A7, we plot the statistics on each categories in order to determine if there is a particular population of crater on which our algorithm doesn't work. It seems that the position of craters identified as "ghost" craters are a little bit more difficult to detect. This is expected because ghost craters are well eroded. As a reminder, to improve the detection of this class of crater, we used a specific image pre-processing as explain in Section 3.2. We also can see that of craters identified as "secondary" craters are sometimes difficult to detect because their shape can significantly differ form a circle.

Appendix B: Comparison with Lagain et al. (2021) Classifications for Age Estimation

B1. Methodology

In this appendix section we perform a comparative analysis of crater-based age estimates using the existing manual classifications from Lagain et al. (2021) with and without discarding the ghost and secondary craters. Also we compare these results with our analysis (detection + classification), that is able to detect and classify crater at smaller size. The goal is to evaluate how our algorithm improves the surface age determinations in the Zhurong landing region (Utopia Planitia), particularly for craters in the 2–10 km diameter range.

We applied the Lagain et al. (2021) catalog to the same region within the 4° × 4° quadrangle analyzed in Subsection 5.2 (Figure 12). Due to the catalog's limitation, only craters ≥1 km were considered. We optionally filter the data set to retain only *Regular* and *Layered* craters, using our YOLO-based classification tool. Both resulting cumulative crater size-frequency distribution (CSFD) were fitted using the Hartmann (2005) chronology model to estimate the surface age. All those results are then compared with our results from Section 5.2.

B2. Results

B2.1. Crater Population and Spatial Distribution

Figure B1 illustrates spatial distribution of craters from the Lagain et al. (2021) catalog (crater diameter ≥1 km), to be compared with Figure 12. Our approach (Martinez et al., 2025b) is able to detect 146,061 crater with diameter ≥1 100 m (see Table 4) whereas only 117 craters are present in the Lagain database (Table B1). Our classification tool identifies 30,252 Regular and 308 Layered craters compared to only 102 Regular +1 Layered in the Lagain catalog (diameter ≥1 km) for this region.

Thanks to our analysis, craters from 100 m to 1 km diameter can be used for a more robust age determination, particularly for younger surfaces where small craters dominate the CSFD.

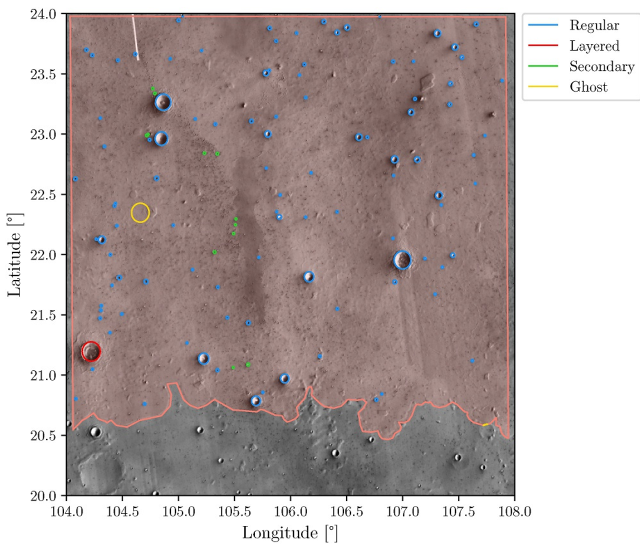

Figure B1. 4° × 4° quadrangle in Utopia Planitia (in equirectangular projection, lower right coordinate: 20°N, 104°E). All craters are referenced in Lagain et al. (2021). Our analysis of the very same area is presented in Figure 12.

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Table B1
Total Number of Craters in the Zhurong Landing Area, Classified Using the Lagain et al. (2021) Catalog

	ø > 1 km	ø > 2 km	ø > 10 km
Regular	102	24	0
Layered	1	1	0
Secondary	12	0	0
Ghost	2	2	0
Total	117	27	0

B2.2. Age Estimation Comparison

Figures left and right 24 present the cumulative crater size frequency distributions (CSFDs) and the derived age estimations for the Lagain database, including all crater classes and only *Regular* + *Layered* craters, respectively. The right plot shows a better agreement with the expected theoretical model (Michael, 2013), especially in the 2–10 km diameter range, demonstrating the necessity to remove the ghost and secondary crater.

By comparing Figure B2 using Lagain database only and our analysis in Figure 13, to the manual reference method of this zone (Wu et al., 2022), we can evaluate the added-value of our approach.

From all the method, our one with classification (Figure 13 right) is the best matching the CSFD from Wu et al. (2022). This can be seen especially observed in the 1–2 km range. The Lagain database, even filtered gives a significantly too large CSFD, with an age (3.28 Gy for filtered data) older than expected.

B2.3. Conclusion

This comparative analysis demonstrates that while the Lagain et al. (2021) catalog provides a foundational data set for Martian crater studies, its limitations in crater size ≥1 km can lead to biased age estimates, at least for this region of Utopia Planitia. Our approach of detection (Martinez et al., 2025b) and then classification (this article), by extending the crater analysis to ≥100 m diameter craters and filtering out non-primary craters, offers a more robust tool for crater-based age determination.

However, we emphasize that this comparison is not intended to undermine the value of manual catalogs. Rather, it illustrates the complementary role of automated methods to refining age estimates, particularly for terrains where small craters are geologically significant.

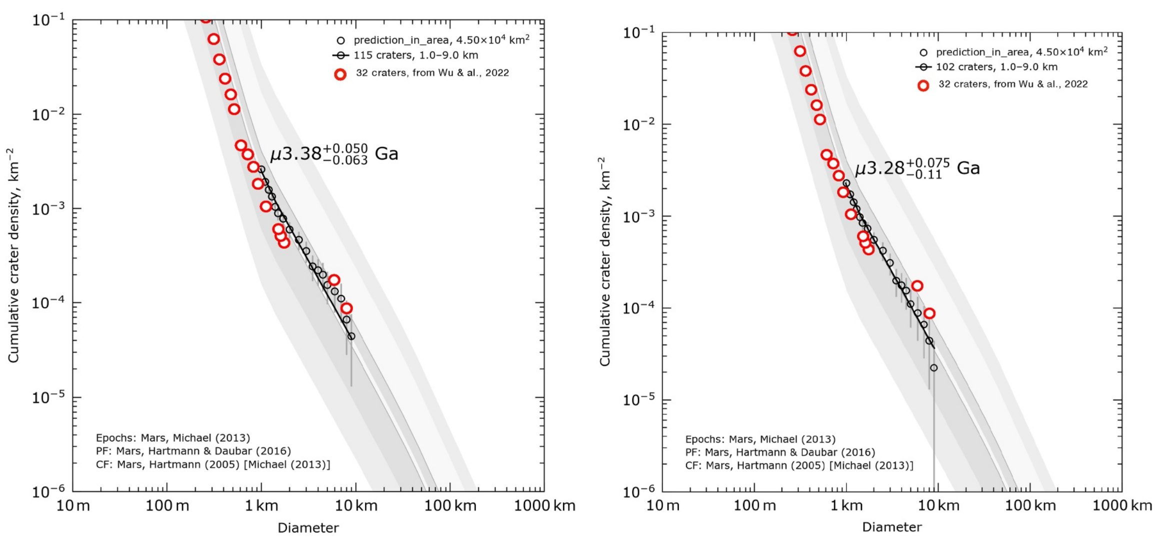

Figure B2. Cumulative crater size frequency (CSFD) of the Lagain et al. (2021) crater database within the Utopia Planitia region (see Figure 12) (left) all crater (right) only considering regular and layered craters determined by the Lagain et al. (2021) crater database (discarding ghost and secondary craters).

Appendix C: Extra Figures

Figure C1 shows the fraction of no data (NaN) within the quadrangles.

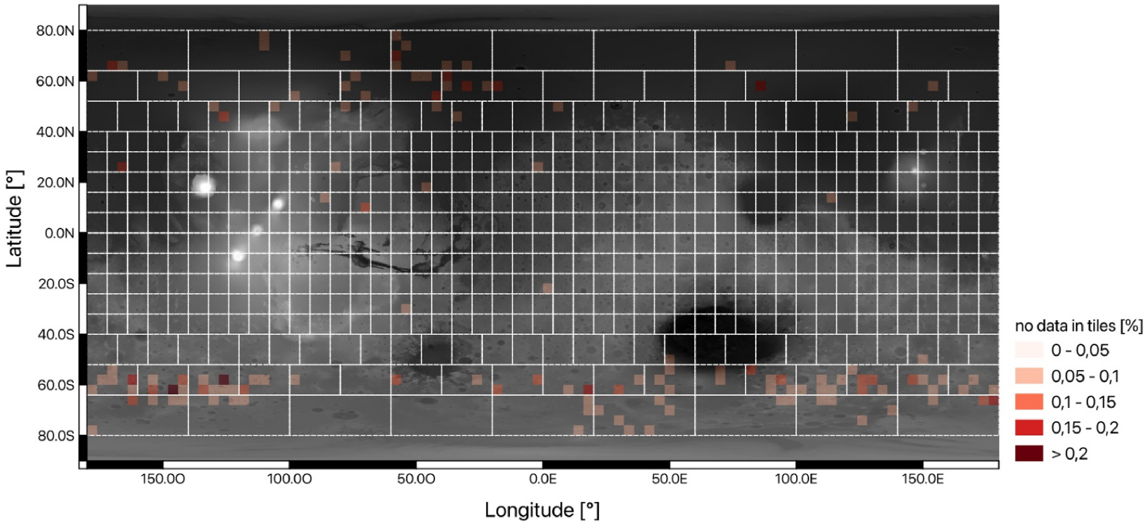

Figure C1. Fraction of no data within the quadrangles.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

To ensure transparency and reproducibility, all data sets, model weights and algorithms used in this study are publicly accessible. Specifically:

- The raw image data set is available online, as described in Dickson et al. (2024). Access the data set here: https://murray-lab.caltech.edu/CTX/tiles/.
- The crater database used for this study is available on GitHub: https://github.com/alagain/martian_crater_database
- The crater detection algorithm is accessible in Martinez et al. (2025a)
- The classification algorithm and model weights are available on GitLab: https://gitlab.dsi.universite-paris-saclay.fr/planet/acdc/pipeline_ACDC

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References

Baker, D. M., Head, J. W., Fassett, C. I., Kadish, S. J., Smith, D. E., Zuber, M. T., & Neumann, G. A. (2011). The transition from complex crater to peak-ring basin on the Moon: New observations from the lunar orbiter laser altimeter (LOLA) instrument. *Icarus*, *214*(2), 377–393. https://doi.org/10.1016/j.icarus.2011.05.030

Barlow, N. G., Boyce, J. M., Costard, F. M., Craddock, R. A., Garvin, J. B., Sakimoto, S. E. H., et al. (2000). Standardizing the nomenclature of martian impact crater ejecta morphologies. *Journal of Geophysical Research*, *102*(E11), 26733–26738. https://doi.org/10.1029/2000JE001258

Benedix, G. K., Lagain, A., Chai, K., Meka, S., Anderson, S., Norman, C., et al. (2020). Deriving surface ages on Mars using automated crater counting. *Earth and Space Science*, *7*(3), e2019EA001005. https://doi.org/10.1029/2019EA001005

Bickel, V. T., Conway, S. J., Tesson, P., Manconi, A., Loew, S., & Mall, U. (2020). Deep learning-driven and detection and mapping and of and rockfalls on Mars. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, *13*, 2831–2841. https://doi.org/10.1109/JSTARS.2020.2991588

Chandnani, M., Herrick, R. R., & Kramer, G. Y. (2019). Geologic analyses of the causes of morphological variations in lunar craters within the simple-to-complex transition. *Journal of Geophysical Research: Planets*, *124*(5), 1238–1265. https://doi.org/10.1029/2018je005729

Chen, M., Liu, D., Qian, K., Li, J., Lei, M., & Zhou, Y. (2018). Lunar crater detection based on terrain analysisand mathematical morphology methodsusing digital elevation models. *IEEE Transactions on Geoscience and Remote Sensing*, *56*(7), 3681–3692. https://doi.org/10.1109/TGRS.2018.2806371

Cintala, M. J. (1977). Martian fresh crater morphology and morphometry-a pre-Viking review. In *Impact and explosion cratering: Planetary and terrestrial implications* (pp. 575–591). Retrieved from https://adsabs.harvard.edu/pdf/1977iecp.symp..575C

Cintala, M. J., & Grieve, R. A. (1998). Scaling impact melting and crater dimensions: Implications for the lunar cratering record. *Meteoritics & Planetary Science*, *33*(4), 889–912. https://doi.org/10.1111/j.1945-5100.1998.tb01695.x

Costard, F. M. (1989). The spatial distribution of volatiles in the martian hydrolithosphere. *Earth, Moon, and Planets*, *45*(3), 265–290. https://doi.org/10.1007/BF00057747

Craddock, R. A., Maxwell, T. A., & Howard, A. D. (1997). Crater morphometry and modification in the Sinus Sabaeus and Margaritifer Sinus regions of Mars. *Journal of Geophysical Research*, *102*(E6), 13321–13340. https://doi.org/10.1029/97JE01084

Croft, S. K. (1985). The scaling of complex craters. *Journal of Geophysical Research*, *90*(S02), C828–C842. https://doi.org/10.1029/JB090iS02p0C828

Dickson, J. L., Ehlmann, B. L., Kerber, L., & Fassett, C. I. (2024). The global context camera (CTX) mosaic of Mars: A product of information-preserving image data processing. *Earth and Space Science*, *11*(7), e2024EA003555. https://doi.org/10.1029/2024ea003555

Duda, R. O., & Hart, P. E. (1972). Use of the Hough transformation to detect lines and curves in pictures. *Communications of the ACM*, *15*(1), 11–15. https://doi.org/10.1145/361237.361242

Gotmare, A., Keskar, N. S., Xiong, C., & Socher, R. (2018). A closer look at deep learning heuristics: Learning rate restarts, warmup and distillation. *arXiv:1810.13243v1*.

Goyal, P., Dollar, P., Girshick, R., Noordhuis, P., Wesolowski, L., Kyrola, A., et al. (2018). Accurate, large minibatch sgd: Training imagenet in 1 hour. *arXiv:1706.02677v2*.

Hartmann, W. K. (2005). Martian cratering 8: Isochron refinement and the chronology of Mars. *Icarus*, *174*(2), 294–320. https://doi.org/10.1016/j.icarus.2004.11.023

Hartmann, W. K., & Neukum, G. (2001). Cratering chronology and the evolution of Mars. *Space Science Reviews*, *96*, 165–194. https://doi.org/10.1007/978-94-017-1035-0_6

He, Z., Wang, K., Fang, T., Su, L., Chen, R., & Fei, X. (2024). Comprehensive performance evaluation of YOLOv11, YOLOv10, YOLOv9, YOLOv8 and YOLOv5 on object detection of power equipment. https://doi.org/10.1109/CCDC65474.2025.11090973

Herrick, R. R., & Lyons, S. N. (1998). Inversion of crater morphometric data to gain insight on the cratering process. *Meteoritics & Planetary Science*, *33*(1), 131–143. https://doi.org/10.1111/j.1945-5100.1998.tb01615.x

Hough, P. V. C. (1962). Method and means for recognizíng complex patterns. United States Patent Office. (US Patent n° 3096654).

Hynek, B. M., & Di Achille, G. (2017). *Geologic map of meridiani planum, Mars*. US Geological Survey Scientific Investigations Map. https://doi.org/10.3133/SIM3356.3356

Jegham, N., Koh, C. Y., Abdelatti, M., & Hendawi, A. (2025). Yolo evolution: A comprehensive benchmark and architectural review of YOLOv12, YOLO11, and their previous versions. *arXiv:2411.00201*.

Johnson, J. M., & Khoshgoftaar, T. M. (2019). Survey on deep learning with class imbalance. *Journal of big data*, *6*(1), 27. https://doi.org/10.1186/s40537-019-0192-5

Kalra, D. S., & Barkeshli, M. (2024). Why warmup the learning rate? Underlying mechanisms and improvements. *Advances in Neural Information Processing Systems*, *37*, 111760–111801.

Kalynn, J., Johnson, C. L., Osinski, G. R., & Barnouin, O. (2013). Topographic characterization of lunar complex craters. *Geophysical Research Letters*, *40*(1), 38–42. https://doi.org/10.1029/2012GL053608

Khanam, R., & Hussain, M. (2024). YOLOv11: An overview of the key architectural enhancements. arXiv: 2410.17725. Retrieved from https://arxiv.org/abs/2410.17725

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. In F. Pereira, C. Burges, L. Bottou, & K. Weinberger (Eds.), *Advances in neural information processing systems* (Vol. 25). Curran Associates, Inc. Retrieved from https://proceedings.neurips.cc/paper_files/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf

Lagain, A., Bouley, S., Baratoux, D., Marmo, C., Costard, F., Delaa, O., et al. (2021). Mars crater and database: A and participative project for the classification of and the morphological characteristics of large martian and craters. *Geological Society of America Special Paper*, *550*(29), 629–645. https://doi.org/10.1130/2021.2550

La Grassa, R., Cremonese, G., Gallo, I., Re, C., & Martellato, E. (2023). Yololens: A deep learning model based on super-resolution to enhance the crater detection of the planetary surfaces. *MDPI Remote Sensing*, *15*(1171), 1171. https://doi.org/10.3390/rs15051171

Lee, C., & Hogan, J. (2021). Automated crater detection with human level performance. *Computers and Geosciences*, *147*, 104645. https://doi.org/10.1016/j.cageo.2020.104645

Leighton, R. B., Murray, B. C., Sharp, R. P., Allen, J. D., & Sloan, R. K. (1965). Mariner iv photography of Mars: Initial results. *Science*, *149*(3684), 627–630. https://doi.org/10.1126/science.149.3684.627

Malin, M. C., Bell, J. F., Cantor, B. A., Caplinger, M. A., Calvin, W. M., Clancy, R. T., et al. (2007). Context camera investigation on board the Mars reconnaissance orbiter. *Journal of Geophysical Research*, *112*(E5). https://doi.org/10.1029/2006JE002808

Martinez, L., Andrieu, F., Schmidt, F., Talbot, H., & Bentley, M. (2025a). Automaticcraterdetector. IPSL data catalog. Retrieved from https://data.ipsl.fr/catalog/srv/eng/catalog.search#/metadata/ae042dbb-37c3-4a98-b72c-02d2a191f53b

Martinez, L., Andrieu, F., Schmidt, F., Talbot, H., & Bentley, M. S. (2025b). Robust automatic crater detection at all latitudes on Mars with deep-learning. *Planetary and Space Science*, *260*, 106053. https://doi.org/10.1016/j.pss.2025.106053

Melosh, H. J. (2011). *Planetary surface processes* (Vol. 13). Cambridge University Press. https://doi.org/10.1017/cbo9780511977848

Michael, G. (2013). Planetary surface dating from crater size–frequency distribution measurements: Multiple resurfacing episodes and differential isochron fitting. *Icarus*, *226*(1), 885–890. https://doi.org/10.1016/j.icarus.2013.07.004

Mills, M., Bickel, V., McEwen, A., & Valantinas, A. (2024). A global dataset of pitted cones on Mars. *Icarus*, *418*, 116145. https://doi.org/10.1016/j.icarus.2024.116145

Neukum, G., & Horn, P. (1976). Effects of lava flows on lunar crater populations. *The Moon*, *15*(3–4), 205–222. https://doi.org/10.1007/BF00562238

Neukum, G., Ivanov, B., & Hartmann, W. (2001). Cratering records in the inner solar system in relation to the lunar reference system. *Chronology and Evolution of Mars*, 55–86. https://doi.org/10.1007/978-94-017-1035-0_3

Parker, M. v. K., Zegers, T., Kneissl, T., Ivanov, B., Foing, B., & Neukum, G. (2010). 3D structure of the Gusev Crater region. *Earth and Planetary Science Letters*, *294*(3–4), 411–423. https://doi.org/10.1016/j.epsl.2010.01.013

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., et al. (2019). Pytorch: An imperative style, high-performance deep learning library. In *Advances in neural information processing systems 32* (pp. 8024–8035). Curran Associates, Inc. Retrieved from https://proceedings.neurips.cc/paper_files/paper/2019/file/bdbca288fee7f92f2bfa9f7012727740-Paper.pdf

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, *12*, 2825–2830. http://jmlr.org/papers/v12/pedregosa11a.html

Pike, R. J. (1980). Formation of complex impact craters: Evidence from Mars and other planets. *Icarus*, *43*(1), 1–19. https://doi.org/10.1016/0019-1035(80)90083-4

Quantin, C., Popova, O., Hartmann, W. K., & Werner, S. C. (2016). Young martian crater Gratteri and its secondary craters. *Journal of Geophysical Research*, *121*(7), 1118–1140. https://doi.org/10.1002/2015JE004864

Quantin-Nataf, C., Carter, J., Mandon, L., Thollot, P., Balme, M., Volat, M., et al. (2021). Oxia planum and the landing site for the ExoMars "Rosalind Franklin" rover mission: Geological context and prelanding interpretation. *Astrobiology*, *21*(3), 345–366. https://doi.org/10.1089/ast.2019.2191

Robbins, S. J., & Hynek, B. M. (2011a). Distant secondary craters from Lyot crater, Mars, and implications for surface ages of planetary bodies. *Geophysical Research Letters*, *38*(L05201). https://doi.org/10.1029/2010GL046450

Robbins, S. J., & Hynek, B. M. (2011b). Secondary crater fields from 24 large primary craters on Mars: Insights into nearby secondary crater production. *Journal of Geophysical Research*, *116*(E10003), E10003. https://doi.org/10.1029/2011JE003820

Robbins, S. J., & Hynek, B. M. (2012). A new global database of Mars impact craters > 1 km: 1. Database creation, properties, and parameters. *Journal of Geophysical Research*, *117*(E05004). https://doi.org/10.1029/2011JE003966

Salamuniccar, G., & Loncaric, S. (2010). Method for crater and detection from and martian and digital topography and data using and gradient and value/orientation and morphometry and vote and analysis and slip tuning and calibration. *IEEE Transactions on Geoscience and Remote Sensing*, *48*(5), 2317–2329. https://doi.org/10.1109/TGRS.2009.2037750

See, J. E., Howe, S. R., Warm, J. S., & Dember, W. N. (1995). Meta-analysis of the sensitivity and decrement in vigilance. *Psychological Bulletin*, *117*(2), 230–249. https://doi.org/10.1037/0033-2909.117.2.230

Shorten, C., & Khoshgoftaar, T. M. (2019). A survey on image data augmentation for deep learning. *Journal of big data*, *6*(1), 1–48. https://doi.org/10.1186/s40537-019-0197-0

Su, S., Fanara, L., Xiao, H., Hauber, E., & Oberst, J. (2023). Detection of detached and ice-fragments at martian and polar and scarps using and a convolutional and neural network. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, *16*, 1728–1739. https://doi.org/10.1109/JSTARS.2023.3238968

Tanaka, K., Robbins, S., Fortezzo, C., Skinner, J., & Hare, T. (2014). The digital global geologic map of Mars: Chronostratigraphic ages, topographic and crater morphologic characteristics, and updated resurfacing history. *Planetary and Space Science*, *95*(mapping), 11–24. https://doi.org/10.1016/j.pss.2013.03.006

Tao, Y., Xiong, S., Conway, S. J., Muller, J.-P., Guimpier, A., Fawdon, P., et al. (2021). Rapid single image-based DTM estimation from Exomars TGO CaSSIS images using generative adversarial U-Nets. *MDPI Remote Sensing*, *13*(15), 2877. https://doi.org/10.3390/rs13152877

Taylor, L., & Nitschke, G. (2018). Improving deep learning with generic data augmentation. In *2018 IEEE symposium series on computational intelligence (SSCI)* (pp. 1542–1547). https://doi.org/10.1109/SSCI.2018.8628742

Ultralytics. (2025). YOLOv11 by ultralytics. Retrieved from https://github.com/ultralytics/ultralytics

Wu, B., Dong, J., Wang, Y., Rao, W., Sun, Z., Li, Z., et al. (2022). Landing site selection and characterization of Tianwen-1 (Zhurong rover) on Mars. *Journal of Geophysical Research: Planets*, *127*(4), e2021JE007137. https://doi.org/10.1029/2021je007137

Zuo, W., Zhang, Z., Li, C., Wang, R., Yu, L., & Geng, L. (2016). Contour-based automatic crater recognition using digital elevation models from Chang'E missions. *Computers and Geosciences*, *97*, 79–88. https://doi.org/10.1016/j.cageo.2016.07.013